Irregular Labor Supply and Coordination in Firms: Experimental Evidence from India

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Abstract

Over the process of development, production increasingly relies on complementarities between inputs. Firms in developing countries face irregular labor supply: uncertainty in whether and when workers arrive at work. Such uncertainty may be especially costly when production requires coordinating labor inputs. This paper tests whether and how irregular labor supply affects firms' ability to coordinate production inputs, and firm outcomes using a randomized experiment with 92 manufacturing firms in India. In the status quo, three in five workers arrive after their firm's start time on average. I incentivize workers at 20% of their daily earnings to arrive by the start time for four weeks. Incentives increase timely arrival from 41% to 59% of worker-days, driven by these late workers. Worker output increases by 24% for all workers in treatment firms, not only those who changed their arrival times as a result of the incentive. I show that these spillovers come from gains along the production line due to improved coordination. Treatment firms produce 29% more output, receive 21% more new orders and are 40% more likely to complete orders on time.

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1 Introduction

Labor supply to firms in developing countries is highly *irregular*. Firms face high rates of absenteeism and separation, indicating irregularities along the extensive margin – workers' choice of which days to work or whether to stay in a firm (Breza and Kaur, 2025). Even when workers are present at work, their labor supply can be irregular along the intensive margin – when they arrive at work, or how many hours they choose to work on a given day. Irregular arrival times have been observed empirically across several settings but are relatively understudied in the literature.[1]

Despite the prevalence of irregular labor supply along both these margins, there is little causal evidence examining whether and how it affects firm outcomes. Irregular labor supply may be especially costly when production relies on complementarities between labor inputs (Kremer, 1993). For example, in assembly lines, one worker's late arrival can hold up other workers on the line. This can have broader implications for economic development as firms increasingly rely on such production processes, and the costs of coordinating workers rise with scale (Lucas, 1978).

How does irregular labor supply affect firms' ability to effectively coordinate production, and firm outcomes? I answer this question in the context of a large garment manufacturing cluster in India comprising small-medium firms. Two features of this setting make it ideal for this study. Firms are organized as production lines where each worker performs a distinct operation on the garment in batches of 80-100 pieces, which are passed sequentially from one worker to the next. This may create complementarities between workers, so that worker output depends on both their own and their co-workers' labor supply decisions. For example, the output of workers performing downstream operations may depend on those performing upstream operations, and therefore their labor supply decisions.

A second feature of this setting is that workers are paid operation specific piece rates, and are neither incentivized based on the number of finished pieces, nor explicitly incentivized to arrive by their firm's start time. As a result, worker arrival times at the firm are highly dispersed, ranging from 3 hours early to 5 hours late, with a 10-90 gap of 135 minutes, more than 2 hours. In fact, arriving after the firm's start time (henceforth referred to as "late arrivals", or "being late") is at least as common as absenteeism: workers are absent on 30% of worker-days, and late on 40% of worker-days conditional on being present. Survey responses suggest that late arrivals reflect important time-commitments outside of work

[1] For example, among casual day laborers (Cefala et al., 2024), government employees (Chaudhury et al., 2006), and factory workers in the setting of this paper.

(such as household and domestic care responsibilities), as opposed to behavioral reasons or tardiness. Anecdotally, workers cite a strong preference for flexible work schedules as the main reason for late arrivals.[2]

These two features of the setting present a puzzle. Sequential, batch processing lines may create possible complementarities in worker arrival times, indicating lateness may be costly to the firm. Indeed at baseline, 90% of owners report they would like their workers to start work at the same time as a worker-related measure to increase their firms' production, and 40% cite late arrivals among the top three challenges faced in running their business. Yet, firms do little to coordinate arrival times: 80% of owners allow workers to be more than 30 minutes late and almost none strictly enforce common start times. Owners correctly attribute this to workers' strong preference for flexibility: 52% believe that enforcing common start times would cause workers to leave the firm or refuse to work there. This could reflect (1) an absence of gains from coordination, (2) a compensating differential for giving up flexible schedules that is too high for firms to pay, or both.

I conduct an experiment with 92 firms comprising 600 workers to test whether reducing lateness generates coordination gains within the firm, and improves firm outcomes. I incentivize workers in 46 randomly selected treatment firms to arrive by a fixed start time for a period of 4 weeks. Treatment incentives are designed as a daily bonus equivalent to 20% of workers' average daily earnings, paid conditional on arriving at the firm by the owners' work start time (hereafter, preferred start time), elicited at baseline. Workers in control firms receive unconditional transfers during the treatment incentive period, matched to a randomly chosen treatment firm each week to ensure that treatment effects are not driven by income effects on worker output.

I collect daily data on worker attendance, arrival times, task assignments and production output for 6-weeks: 2 weeks before the incentives, and the 4-week incentive period. This allows me to estimate the effect of the incentives on labor supply (timely arrival, attendance and daily work hours) and production outcomes (worker and firm output, earnings, and order-completion delays) using a difference-in-differences specification with worker and time fixed effects.

This design allows me to address both explanations of this puzzle. Comparing production outcomes between treatment and control firms directly tests for coordination gains as a result

[2]This is consistent with prior work documenting that workers in developing countries have a greater demand for flexibility, given the prevalence of health shocks (Breza and Kaur, 2025), social pressure to attend cultural/family events (Goraya et al., 2025), and gender norms around domestic responsibilities (Ho et al., 2024). Consequently, late arrivals are more common among women.

of timely arrival, the primary goal of this paper. In addition to this, workers' labor supply response to the incentives reveals how much they value flexible arrival times. Treatment firm workers who take up the incentive value flexibility at no more than 20% of daily earnings, while those who don't value it at more than 20%. To maximize power for the first test, I chose a single, large incentive amount. The amount was decided based on pre-piloting discussions with firm owners about the minimum bonus amount that would shift worker arrival times at the firm. The incentive is therefore costly by design, and it is not intended to be a profit-maximizing contract for firms.

I find that the incentives successfully encourage workers to arrive on time in treatment firms. Timely arrival increases by 43% relative to control firms (18 percentage points). This translates to 1 additional worker out of 5 workers on average being present at the firm before the preferred start time. Further, workers change their labor supply entirely along the intensive margin by arriving earlier at the firm. There is no change in attendance rates for the average worker, and a small but imprecise increase in daily work hours that partly offsets the amount by which workers were late at baseline.[3] To identify who responded most to treatment, I classify workers as "Early", "Late", and "Very Late" based on their average arrival times during pre-incentive weeks. The timely arrival effect is driven entirely by "Late" and "Very Late" workers, on whom the incentive was most binding: "Late" ("Very Late") workers are 54% (21 p.p) and 88% (35 p.p) more likely to arrive on time, and thus work 20 minutes and 43 minutes more per day compared to the average control worker. Finally, I find no evidence that "Early" workers altered their labor supply in response to the incentives.

This increase in timely arrival in treatment firms results in a meaningful decline in dispersion in arrival times. Total dispersion in worker arrival times, measured as the difference in arrival times between the earliest and the last worker, decreases by 22 minutes (17% of the control mean). This is driven by the last worker arriving on average 14 minutes earlier in treatment firms, a 29% reduction relative to the control mean of 47 minutes. Incentives thus improve coordination in worker arrival times by shifting the arrival time distribution to the left.

As a result, weekly output per worker in treatment firms increases by 0.26 standard deviations, a 24% increase relative to the control mean (equivalent to 930 additional pieces per week).[4] I also find suggestive evidence that weekly worker earnings increase by 9%,

[3]Workers in treatment firms work on average 17 minutes longer per day than control firm workers, a 5% increase relative to the control mean. The average worker arrives 8 minutes late during pre-incentive weeks.

[4]Worker output sums the number of pieces stitched across all operations in a week, and is standardized to pre-incentive, product-level means.

though the estimate is imprecise. The increase in output per worker partly reflects an increase in per-hour worker productivity: the effect on output survives even after scaling by weekly hours worked, but is smaller. I provide suggestive evidence that this increase stems from more productive time utilization among "Late" workers: "Late" workers in treatment firms are 4 percentage points less likely to perform non-production tasks (a 25% reduction relative to control workers), and instead spend more time during the work day on tailoring tasks.[5]

Crucially, the incentives generated positive spillovers among workers in treatment firms. The increase in worker output is realized by all workers, including both "Early" workers who did not respond to incentives, and those who arrived late at baseline and responded most to the incentives. Gains in worker earnings and output per hour follow a similar pattern of spillovers from "Late" to "Early" workers. The same pattern holds by gender: women are more likely than men to be "Late" workers, and their labor supply response is twice as large as their male co-workers. Yet, men and women in treatment firms gain equally in weekly output, suggesting similar spillovers from female to male co-workers.

These spillovers reflect gains from improved coordination along the production line. I classify firms as "Bottleneck" firms if any worker whose modal baseline task is upstream in the production line (an upstream task worker) is a "Late" worker at baseline; remaining firms are classified as "Non-Bottleneck" firms, where no upstream task workers are "Late". If there are gains from coordination, I expect to see output gains among "Early" workers to be driven by Bottleneck firms, where lateness creates hold-ups for other workers. In Non-Bottleneck firms, lateness does not create hold-ups for other workers, but may result in delays in order-completion timelines. These definitions use pre-incentive data, and treatment does not predict Bottleneck status (p-value = 0.42). Treatment and control firms are also balanced on baseline characteristics within each sub-sample of firms.

Consistent with the prediction above, I find that the effect on weekly output for the average worker is larger in Bottleneck firms (0.36 SD) than in Non-Bottleneck firms (0.12 SD, not statistically significant). More importantly, *which* workers experience this gain differs across the two samples. I find that spillovers from "Late" to "Early" workers only exist in Bottleneck firms: both "Early" and "Late" workers experience a 0.39 SD and 0.34 SD increase in weekly output respectively, and these effects are not statistically distinguishable from each other. Adding further support for the coordination channel, I find that in Non-Bottleneck firms, gains in worker output are concentrated only among "Late" workers, with

[5]At endline, treatment firm owners are more likely to report a reduction in idle time among workers, providing anecdotal support that coordinating worker arrival times improves time utilization in the firm.

no increase in output for "Early" workers. Gains in earnings and output per hour, and improvements in time utilization are more pronounced in Bottleneck firms, consistent with improved coordination along the production line. Since Bottleneck and Non-Bottleneck status are not randomly assigned, the two sub-samples differ on some baseline characteristics. In particular, Bottleneck firms are larger, have more "Late" workers, and more upstream task workers who are "Late". These differences simply reflect how Bottleneck firms are defined, and can produce larger effects on output through a larger labor supply response. However, they are less likely to explain the presence of spillovers to "Early" workers. Additionally, these patterns are robust to controlling for the number of "Late" workers in an ANCOVA specification, as well as an alternate definition of hold-ups within the firm.

These gains in worker output translate almost one-to-one to gains in total firm output.[6] I find a 0.30 standard deviation increase in weekly firm output: a 29% increase relative to the control mean (equivalent to $\approx$ 700 pieces). Similar to worker level outcomes, gains in total firm output are larger in Bottleneck firms, suggesting that the firm also stands to gain from improving coordination in worker arrival times.

Additionally, treatment facilitates timely completion of orders. Treatment reduces the probability of completing an order after the owner's planned delivery date by 10 percentage points, a 40% reduction relative to the control mean. As predicted, in contrast to effects on total output, this reduction in delays is driven by Non-Bottleneck firms: although lateness occurring downstream in the production line does not create an externality on other workers, it can hold up the delivery of finished pieces. Timely completion of orders is crucial in this setting because most firms rely on repeat relationships with their clients to procure new orders. Consistent with this, I find suggestive evidence that treatment firms receive more new orders: the quantity of new orders received during the incentive weeks increases by 0.27 SD, a 21% increase relative to control firms ($\approx$ 880 pieces).

Finally, I find that weekly firm revenues increase by INR 3,257 (8% relative to the control mean), while total labor costs, *excluding* the incentive paid to treatment workers, increase by INR 1,640 (9% relative to the control mean). Weekly firm profits, excluding incentive payments made to workers, increased by $\approx$ INR 1,600 (8% of the control mean), suggesting irregular labor supply may reduce firm profits, although the effect is noisy. However, paying a 20% bonus as incentives for timely arrival outweighs revenue gains, resulting in a negative effect on profits of $\approx$ INR 981. The incentive was large by design: during pre-piloting work, owners suggested roughly 20% of earnings was what it would take to shift arrival times. This

[6] Total firm output is measured as the total number of finished pieces produced by all workers in the firm, and standardized to pre-incentive, product-level means.

is consistent with workers placing a high value on flexible arrival times, thus making it too costly for firms to incentivize timely arrival despite the gains.

Anecdotally, treatment firm owners were 26 percentage points (96%) more likely to agree that total production at their firm increased during the incentive weeks, and were twice as likely to attribute this increase to workers arriving on time relative to control firm owners. This indicates that owners found the incentives valuable. However, given the small gains in profits (excluding incentives), or the costs of incentivizing timely arrival, it is perhaps not surprising that only 26% of treatment firm owners were willing to continue paying for incentives themselves at the end of the experiment. Owner's average *stated* willingness to pay was INR 30 – less than a third of the incentives that I offered. In a 3-12 month follow-up survey, there was no take-up of any kind of financial incentives among treatment firms, beyond verbal encouragement.

This lack of enforcement, and low willingness to contract on coordinated arrival times may reflect market-wide norms: firms operate in dense clusters with an active market for experienced tailors, where nearly all firms pay tailors piece rates and don't enforce common arrival times. This makes it difficult for a single firm to deviate without facing difficulties in hiring or retaining experienced tailors. This paper shows that returns to coordination within the firm are large and economically meaningful. However, given that drivers of lateness are related to poor quality of infrastructure and norms around domestic and caregiving work, firm-level incentives alone may not be a sustainable solution.

This paper is most closely related to the literature on barriers to firm growth in developing countries. I extend the literature examining the impact of irregular labor supply on firm outcomes (Krishnaswamy, 2019; Goraya et al., 2025) in two key ways. First, I provide causal evidence that firms gain from timely worker arrival through coordination gains along the production line using high frequency data on worker labor supply and task assignments within the firm. This extends the finding from previous work that adverse effects of worker absenteeism are stronger in firms and industries with larger co-worker complementarities. Second, I show that timely arrival is a meaningful driver of worker and firm outcomes in developing countries. This paper is closest to the work studying the effect of attendance incentives among government workers in developing countries (Banerjee et al., 2008; Duflo et al., 2012; Dhaliwal and Hanna, 2017). While these studies used time-stamped monitoring to verify worker attendance, their focus was exclusively on attendance and not timely arrival. To the best of my knowledge, this is the first paper to experimentally test the effects of timely arrival on firm outcomes in the private sector.

A large body of work documents search and hiring frictions in low-income countries (De Mel et al., 2019; Hardy and McCasland, 2023; Bassi and Nansamba, 2022; Caria and Falco, 2024; Carranza et al., 2020). I add to this work by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. Specifically, I show that a lack of coordination in worker arrival times affects both workers' and firms' ability to realize the returns from coordination in a team production setting. To the extent that coordination becomes costlier with scale (Lucas, 1978), this can have important implications for firms' ability to scale their workforce. Additionally, I highlight worker preferences for flexibility as a novel barrier faced by developing country firms in coordinating labor inputs.[7]

I also contribute to the growing literature on the internal organization of small, informal firms in developing countries, building on recent work such as Bassi et al. (2022, 2023). In particular, Bassi et al. (2023) find that limited labor specialization in multi-worker Ugandan manufacturing firms limits firms' ability to organize labor productively. Instead, these firms are organized as a collection of self-employed individuals, and are unable to realize gains from team production. By contrast, I study a setting with labor specialization along a sequential production line and provide evidence for coordination gains within firms.

Finally, this paper speaks to work on the provision of flexible work arrangements by firms and labor supply preferences (Mas and Pallais, 2020). Workers' demand for flexibility has been documented across several different contexts (Mas and Pallais, 2017; Ho et al., 2024; Baysan et al., 2024; Wiswall and Zafar, 2018). I add to this work by showing that while workers prefer flexible work arrangements, their own labor supply decisions create externalities on their co-workers. This is consistent with the idea that despite fixed-schedules being a dis-amenity for workers individually, there are potential gains to themselves *and* the firm.

The remainder of the paper is organized as follows: section 2 describes the study context and data, section 3 describes the experimental design and implementation, and section 4 lays out the empirical strategy. section 5 presents the main results and section 6 concludes.

[7] As recent work highlights, such preferences may be a consequence of under-development itself, owing to the prevalence of health shocks, poor infrastructure reliability, kinship pressure, etc. (Breza and Kaur, 2025).

2 Setting, Data and Descriptive Evidence

2.1 Setting

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units, that specialize in distinct stages of the supply chain, such as knitting, dyeing, cutting, stitching, and packing. This paper focuses on job-work units that specialize in stitching, the most labor-intensive stage of production. Each firm is a *small* factory that receives cut pieces of fabric from a client/buyer (hereafter referred to as an order), and is responsible for stitching the finished garment and delivering the order back to their client. Firms specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,000 pieces, varying with product type.

Production is organized in batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations (hereafter, tasks) need to be performed sequentially in order to produce a finished piece of garment. The main production workers in these firms, and the focus of the study, are tailors, who perform designated tasks along the production line. Figure 1 depicts a stylized production line for a firm producing T-Shirts. In this example, the firm receives cut pieces of fabric. One tailor first attaches the shoulders, passing on a batch to a second tailor who stitches the sleeves, followed by a third tailor who attaches the collar, and finally the fourth tailor who sews the bottom hem. Firms also hire non-production workers called helpers. Helpers are unskilled workers whose primary role is to support the production line by preparing the fabric and moving work-in-progress inventory to the next substation. While helpers are not the focus of the study, I utilize the presence of production and non-production tasks in the production process to measure tailor's time-utilization within the work day.

The study takes place in 4 urban and peri-urban neighborhoods in Tirupur which were identified as dense clusters of small-medium stitching firms during prior scoping and area mapping exercises. I target firms employing between 5-20 workers, with at least 2 tailors who are not related to the firm owner or those who work for pay. All firms that satisfy these criteria were eligible for the experiment. [8]

[8] While these firms are larger than modal, small developing country firm, they employ $\approx 30\%$ of the workforce within the garment industry in Tiruppur (Economic Census, 2013-14). Such organization structures

2.2 Data

This section outlines the data collected from firms and workers as part of the experiment:

Baseline: I conduct a baseline survey with owners to capture owner and firm characteristics, and a complete worker roster, which is followed by a baseline survey of all workers at the firm. Randomization and treatment announcement was done after owner and worker baseline surveys.

Baseline firm and worker characteristics, and balance by treatment arms are reported in Table 1 and Table 2 respectively. On average, firms hire 6 tailors (excluding owners/co-owners), and have been operating for 6-7 years. Firms produce 4,000 pieces of output each week, and earned a profit of INR 30,000 in the last 30 days (USD 340).

Owners directly supervise workers (that is, without a manager/contractor) in almost all firms – owners also perform stitching work in 87% of firms, and work as regular tailors in about 43% of firms. About 85% owners are male, 57% have completed secondary school and have 19 years of experience in the garment industry.

Workers are on average 37 years old (similar to owners), 44% are male and most workers are married (94%) and have at least one child younger than 14 years of age (85%). 87% of workers live within a 15 minute commute to the firm, have worked at the current firm for 2.5 years and have 15 years of experience in the garment industry. Both firm and worker characteristics are balanced across treatment and control firms (Table 2).

Order, Task and Piece Rates Data: At baseline, I also record a roster of all distinct styles that the firm is currently producing. For each style, I record the piece rate that the owner receives for a finished stitched piece and its task structure – a list of all tasks that need to be performed to produce the finished product, the order in which they are performed, and task specific piece rates paid to workers. This roster is updated throughout the 6-week study period to incorporate new styles that the firm is producing.

Daily Production and Attendance Data: Starting 1-2 weeks after baseline, I collect data *daily* for a period of 6-weeks, out of which workers are paid incentives for 4-weeks.[9] Each day, I record workers' attendance, arrival time, break times during the day, and daily output

are common in many industrial clusters around the world, for example Nadvi and Schmitz (1994); Banerjee and Munshi (2004); Chaudhry and Woodruff (2013); Atkin et al. (2017).

[9]In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I collected 2 weeks of pre-incentives data only, dropping the post-incentive data collection. In Phase 3, I collected 2 weeks of pre-incentives data, 4 weeks of incentives data and 1 week of post-incentives data.

(by product-task). Arrival times and attendance are recorded by surveyor observation and are not prone to mis-reporting in order to game the incentive; remaining data is self-reported by workers. Daily output is recorded for all production workers, including firm owner(s), helpers, and temporary workers where applicable. I use daily output data from all production workers to construct a measure of total firm production. In addition to tailoring/production output, I also record whether the worker did any other work at the firm, such as helper work, receiving or delivering raw material/finished pieces, machine repair, purchasing inputs, etc.

Orders Data: At the end of each week, I record all orders received and delivered by the firm in the past week. This allows me to track new orders received by firms, defined by new batches of cut-material received, and order completion timelines. Order completion timelines are based on owners' reported planned versus actual completion dates per order, allowing me to construct a measure of production delays.

Endline: I conduct endline surveys during the last week of the study to capture firm owners' and workers' perceptions of the incentives, preferences for flexible work arrangements, changes in firm operations during the study and time-use during and outside of work (relative to the pre-study period). These questions allow me to examine (1) the nature of workers' commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers' preferences for flexible (versus strict) work arrangements.

Follow-up: Finally, I conduct a follow-up survey ranging 3-12 months after the experiment to record the types of measures firms adopted to encourage timely arrival, and long-term firm outcomes (survival, temporary closures, business finances and worker turnover).

2.3 Descriptive Evidence on Worker Arrival Times

Tailors are paid piece rates for each individual task they perform (for example, attaching shoulders), and not the number of finished pieces of the garment. Although firms have set operating hours, tailors are not explicitly incentivized to arrive at a common start time. As a result, despite the sequential nature of production, the pay contract does not give tailors any private incentive to coordinate their arrival times. Consequently, I observe substantial dispersion in worker arrival times around the owners' preferred start time, as shown in Figure 2: the 10-90 gap of this distribution is 135 minutes – just over 2 hours. For the remainder of the paper, I define late (timely) arrivals as arriving after (on or before) the preferred start time. I use average pre-incentive arrival times to classify workers into three

types: "Early", "Late", and "Very Late" workers. "Late" and "Very Late" are indicators for workers whose average pre-incentive arrival times were between 0 and 30 minutes, and above 30 minutes of the preferred start time, respectively.

Lateness is both persistent and unpredictable. Persistence in timely-arrival across two consecutive workdays for the average worker is 0.55 (SE = 0.04, p-value < 0.001), but falls to 0.06 (SE = 0.041, p-value = 0.175) after accounting for worker fixed effects. About half the variation in timely-arrival is within worker (intra-class correlation = 0.46), suggesting that lateness is a characteristic of workers that can be learned by owners. This is consistent with focus group discussions and survey data from firm owners about which workers at the firm are typically late, and their usual arrival times. However, workers often deviate from their own arrival time norms, making it difficult for owners to predict actual arrival times. For example, owners could correctly predict which workers will arrive on time the next day 63% of the time, and which workers will arrive on time for the entire week only 23% of the time.[10] On the other hand, owners are better at predicting worker attendance: they correctly predict who will be present the next day 87% of the time, and for the entire week 57% of the time. This motivates understanding the effects of late arrivals on firm outcomes in addition to worker absences.

2.4 How Lateness Affects Production

Lateness can affect production both directly and indirectly. Pre-incentive data on daily work hours and output suggests that "Late" and "Very Late" workers work fewer hours, but don't produce more pieces per hour relative to "Early" workers. Thus, regardless of other workers' arrival times, "Late" and "Very Late" workers may result in lower firm output through the direct channel of producing fewer pieces of output themselves.

However, given the sequential nature of production, late arrivals may also have indirect effects on other workers in the firm. Specifically, the lack of coordination in arrival times and the presence of both "Early" and "Late"/"Very Late" workers in the firm can create bottlenecks in production in two ways. First, lateness on upstream task stations in the production line can hold-up workers on downstream tasks by either increasing idle time – if "Early" workers wait for the "Late" worker to complete upstream tasks, or limiting downstream workers' task specialization – if "Early" workers need to fill in for the upstream task. Both outcomes are less desirable to workers given workers are not paid on time,

[10] Owners are also over-optimistic about timely arrival, and more so for "Late" workers: owners reported 52% of their workers will be on time for the entire week, compared to actual on-time rate of 41%. This gap is larger among workers who are late on average: 43% predicted versus 23% actually on time.

and workers prefer to specialize in their primary tasks. On the other hand, lateness on downstream task stations doesn't create externalities on other workers on the production line. However, it may hold-up the delivery of finished garments, and thus create delays in order completion.

This framework is consistent with owners' beliefs about how lateness affects production at their firm. At baseline, 90% of owners report that they would like their workers to start work at the same time as a worker-management measure that would help increase their firm's production.[11] When asked why coordinated timings would help increase production, 66% of owners suggest a reduction in idle time, and 51% suggest increased task specialization as possible mechanisms (Figure 3b). Additionally, 40% owners also allude to uncertainty about whether and when workers will arrive at the firm as another mechanism. Coordinated timings can help owners plan and deliver orders on time, which helps them receive more orders in the future.

2.4.1 Adaptation

In theory, the firm can adapt to lateness in several ways. First, owners can simply enforce common arrival times, either by penalizing, or firing workers who arrive late. However, firms are located in dense clusters and surrounded by several other firms operating in a similar fashion: almost all firms pay tailors task specific piece-rates, and most owners allow workers to arrive late. Given the market equilibrium, owners are concerned that enforcing common arrival times would result in workers leaving the firm, possibly harming their reputation in the market and making future hiring difficult.

Firms can re-organize production to adapt to lateness, but their ability to do so may be limited. For instance, firms cannot change the sequence in which tasks need to be performed, as the sequence depends on how fabric is cut, a decision made by their clients. Firms can instead re-assign workers to tasks such that workers who arrive early are assigned to upstream tasks, and those who arrive late are assigned to downstream tasks. However, worker-tasks assignments are somewhat fixed, in that workers specialize in their main task. Figure A4 shows that 77% of workers have one main task at the firm, defined as the modal task they worked on during pre-incentive weeks. Given day-to-day fluctuations in absenteeism and lateness, there is some task-sharing observed at baseline: Figure A5 shows that on average, workers produce 71% of the day's output from their main task, and the remaining on other

[11] This far outweighs other measures – 38% of owners would like workers to take fewer/shorter breaks, 25% would like workers to take fewer un-announced leaves and 17% to not bring their children to work (Figure 3a).

tasks. However, conversations with workers suggest that task-sharing is not preferred for two reasons. First, workers recognize there are gains from specializing in a single task relative to performing multiple tasks. Second, there are informal norms against task sharing, particularly when tasks differ in piece-rates: a worker who typically performs a high piece-rate task might dislike her work being covered by a co-worker when she is late. Figure A6 also suggests there is no correlation between whether a person is late and the sequence of the task they perform at the firm, consistent with task-assignments not changing in response to daily fluctuations in arrival times. Finally, task-sharing might not always be possible in practice if skills are not transferrable across tasks. For example, stitching tasks

The main way in which firms do adapt to lateness is by maintaining work-in-progress inventory between work-stations. This is a crucial feature of batch processing lines that allows firms to adapt to both worker absenteeism and lateness more flexibly compared to assembly lines (as discussed in Chaudhry and Woodruff (2013)). Maintaining a sufficient pile of material for workers who arrive early may mitigate indirect effects of lateness. While I do not have a direct measure of inventory levels within the day, data on weekly orders allows me to identify the number of distinct orders that are in production each day, an alternate measure of inventory. I observe firms having single orders in production on about half of firm-days, allowing me to assess the extent to which firms successfully adapt to lateness. Pooling all days from pre-incentive weeks and control firms, I find that output and earnings for workers on downstream tasks is lower on days when the upstream task worker is late, and the correlation is much more pronounced on days when the firm has only one active order in production (Table A11). This suggests that firms are only able to partly mitigate indirect effects from lateness.

2.4.2 Two Definitions of Hold-Ups Created by Late Arrivals

The sequential nature of production, and limited skill transferability across tasks allows me to define hold-ups created by lateness in two ways:

1. Bottlenecks created by lateness on upstream tasks

The first definition is based on where lateness occurs on the production line. I define a "Bottleneck" firm as a firm with at least 1 upstream task worker who is also "Late" or "Very Late" at baseline. An upstream task worker is a worker whose main task (as defined at baseline) is an early sequence task in the production line. The complement to this is a "Non-Bottleneck" firm, where no upstream task worker is late. Since firms produce different products, and task sequences may vary across firms, I identify upstream (downstream) tasks

as all tasks that appear before (after) a "cut-off" task. Upstream task cut-offs are defined based on tasks that must be done to proceed with remaining tasks (e.g., the task 'attaching shoulders' must be done before all other tasks to make a t-shirt). Likewise, downstream task cut-offs are defined based on tasks that must be done in order for a garment to be considered finished (e.g., bottom hems on a t-shirt, or elastic bands on pants).[12]

This definition yields two predictions:

1. If lateness creates negative spillovers along the production line, reducing lateness would have both direct and indirect effects on worker output. Spillovers should only be present in Bottleneck firms, and not in Non-Bottleneck firms.

2. If lateness does not create negative spillovers along the production line, reducing lateness would only have direct effects on worker output. However, reducing lateness may help reduce order-completion delays.

2. Spillovers are more likely when tasks sharing is limited

The second definition leverages limited substitutability of worker skills across tasks in the production line. I classify tasks within a firm into two types: Simple and Complex tasks. Complex tasks are tasks performed on a singer machine (as opposed to Overlock or Flatlock machines for Simple tasks), and completing the task for a single piece involves more precision and time. As a result, complex tasks take double the time to complete, are paid a higher piece rates, and only performed by experienced tailors skilled at using the singer machine. Simple tasks can be performed by all tailors, however, Complex task tailors dislike performing lower piece-rate tasks. As a result, there is little crossover between both types of tasks. Further, Complex tasks tailors are significantly more likely to arrive early at the firm than Simple task tailors (possibly because higher piece-rates incentivize them to work longer hours). This means that

I classify firms as "Complex Task" firms if at least one product they produce at baseline has a complex task, and "No Complex Task" firms otherwise (44 and 47 firms respectively). Among Complex Task firms, I further distinguish firms where the complex task follows a simpler task in the production line ("Complex Downstream", 27 firms) from the remaining firms ("Complex Upstream", 17 firms). An advantage of this definition is that unlike the

[12] Figure A3 shows that my classification of upstream and downstream tasks maps well to relative task orders in the production line.

Bottleneck classification, it is not mechanically related to the extent of lateness within the firm at baseline.

This definition yields one prediction: No Complex Task firms are better able to adapt to lateness, since workers can be easily re-assigned to other Simple tasks. On the other hand, lateness may create negative spillovers in Complex Task firms since workers cannot easily be re-assigned across Simple/Complex tasks. Further, since lateness is more likely to occur on Simple task stations, negative spillovers should be more likely to occur in Complex Downstream firms. Spillovers should only be present in Complex Task and Complex Downstream firms, and not in No Complex Task firms.

Finally, the two definitions are not strongly or significantly correlated with each other, and serve as robustness checks for each other.

3 Experiment Design and Implementation

Firms are randomly assigned to one of two treatment groups - (T) Treatment (n = 46), and (C) Control (n = 46). All workers in treatment firms were eligible to receive a daily bonus of INR 100-150 (equivalent to 20% of their *average* daily earnings) if they arrived at the firm and started work by the company's preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings.[13] The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected treatment firm, capped at INR. 300-450 per week (half of the maximum incentive amount in treatment firms). These treatment-control firm (randomized) pairings were created separately each week. section 5 shows the experiment design.

The experiment was implemented in three phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 40 firms (19 treatment, 20 control) from March 2025 to May 2025. Phase 3 was implemented with 35 firms (18 treatment, 18 control) from August 2025 to October 2025. section 5 shows the study timeline. The daily treatment incentive amount in Phase 1 and Phase 2 was set to INR 100, and was increased to INR 150 in Phase 3.[14]

[13] All workers who were working full-time at the firm at any point during the study were eligible to receive incentives. Workers' full-time status was determined by the owner and verified by the research team using daily attendance data.

[14] Follow-up surveys were conducted between November 2025 and January 2026, and were conducted in the order that firms were recruited. Thus, the follow-up timeline ranges from 3 months for Phase 3 firms, to 12 and 9 months for Phase 1 and 2 firms respectively.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, which firms were matched to in the order in which their baseline survey was completed. Randomization for Phases 2 and 3 was stratified on the degree of dispersion in worker arrival times around the owner's preferred work start time, using worker's usual arrival times as reported by the owner at baseline. All specifications reported in the paper control for phase by strata fixed effects to account for the method of randomization in each phase (Bruhn and McKenzie, 2009).

4 Empirical Strategy

I estimate the effect of incentives on worker level outcomes using the following specification:

$$Y_{i,f,t} = \alpha + \boldsymbol{\beta} T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \lambda_{i,f} + \mu_w + \mu_{dow} + \varepsilon_{i,f,t} \tag{1}$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t, and time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time), $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects. μ_w are study-week fixed effects, [15] and μ_{dow} are day-of-week fixed effects (included only in day-level specifications). $\boldsymbol{\beta}$ is the coefficient of interest, and standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level.

I use a similar specification for firm-level outcomes:

$$Y_{f,t} = \alpha + \boldsymbol{\beta} T_f \times \mathbb{I}\{\text{During}\}_t + \lambda_f + \mu_w + \mu_{dow} + \varepsilon_{f,t} \tag{2}$$

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered at the firm level. Some firm outcomes are only measured at endline, in which case I estimate Equation 2 without the t subscript and without firm and study-week fixed effects, and add the baseline value of Y_f as a control. Day-of-week fixed effects are only included in firm-day

[15] Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across all phases. The first pre-incentive week in Phases 2 and 3 is coded week -1, and the final, post-incentive week in Phases 1 and 3 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

level regressions.

5 Results

5.1 Labor Supply

First, I test the effect of the incentives on timely arrival, daily work hours and attendance, reported in Table 3. Timely arrival is defined as an indicator for being present at the firm, *and* arriving by the owner's preferred start time. Daily work hours are defined as the total work duration net of the total duration of breaks taken during the day, and coded 0 when workers are absent.[16] Attendance is an indicator for whether the worker was present at the firm that day. Timely arrival in treatment firms increases by 18 percentage points (column (1)), representing a 43% increase in compliance relative to the pre-incentive control mean (41 percentage points). Column (3) reports the treatment effect on daily work hours. Workers in treatment firms worked an additional 0.28 hours (17 minutes) relative to workers in control firms, but this effect is not significant at conventional levels. Finally, I find no effect on attendance (column (5)).

These results suggest the labor supply response is driven by workers arriving on time, rather than a change in worker attendance. Figure 7 plots the distribution of worker arrival times (conditional on being present) in 15-minute intervals around the preferred start time for treatment (solid lines) and control (dashed lines) firms, visualizing the treatment effect on compliance. Incentives increase the fraction of workers that have arrived within 15 minutes of the preferred start time by 20 percentage points, over a base of 60% in control firms. Given the average size of 6 tailors per firm and 71% attendance rate (4 out of 6 workers present on average), this means that treatment increases number of tailors at the firm by the first 15 minutes of the preferred starting time by 1 additional worker.

Next, I examine whether these effects vary by baseline lateness. Even columns in Table 3 report heterogeneity for "Late" and "Very Late" workers for the main labor supply outcomes. On the other hand, the effect of timely arrival is driven entirely by workers who were consistently late at baseline (column (2)). "Late" and "Very Late" workers are 21 and 35 percentage points respectively more likely than "Early" workers to arrive on time. This is expected, since the incentives were more binding on workers who consistently arrived after the preferred start-time. Finally, The increase in work hours (reported in column (4)) is

[16] Breaks include any time taken for lunch, tea, or other reasons for which the worker left the firm. Workers rarely do other non-tailoring tasks for the firm during breaks (5% of worker-days).

larger among "Late" workers (0.37 hours/22 mins) and "Very Late" (0.73 hours/44 mins) workers, but not statistically significant at conventional levels.

The magnitude of the increase in work hours – 22-44 minutes for "Late" and "Very Late" workers is largely driven by these subgroups of workers arriving earlier than their pre-incentive start times, as opposed to taking longer breaks during the workday or leaving the firm earlier. Table A2 reports the treatment effect on deviation from the preferred start time. On average, workers in treatment firms arrived 11 minutes earlier relative to control workers. This effect is driven by "Late" and "Very Late" workers (column (4)) arriving 13 minutes and 28 minutes earlier relative to control workers.[17]

Treatment effects on attendance for "Late" and "Very Late" workers are not significantly different from those of "Early" workers, consistent with the result above that incentives did not change worker attendance in treatment firms (column (6)). Surprisingly, attendance among "Early" workers does not improve either, given that these workers were effectively only incentivized to be present at the firm without changing their arrival time.[19]

Importantly these results suggest that workers' labor supply response operates entirely along the intensive margin of choosing when to arrive at work. This is supported by the lack of treatment effects on attendance, and that the change in work hours approximately offsets the amount by which workers were late at baseline.

5.1.1 Does treatment improve worker coordination in arrival times?

Next, I examine how treatment affects worker coordination at the *firm-day level*, conditional on being present. Table A5 reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (column (1)), the 90-10 gap in worker arrival times (column (2)), and the deviation from preferred start time for the 90th percentile worker (columns (3)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion

[17] There is some evidence that "Very Late" workers compensate for arriving earlier by taking one additional break during the day, resulting in an 8 minute increase in the total duration of breaks taken during the day (pval = 0.10), as reported in columns (2) and (4) of Table A4. Finally, Table A3 suggests there is no evidence that workers leave the firm before 6.30[18] (columns (1)-(2)), before the firm's usual closing time (columns (3)-(4)), or before their own usual exit time (columns (5)-(6)).

[19] 37% of worker absences are because of social functions, followed by 19% because there was no work at the firm, 18% because either the worker or a family member was sick, and 10% because of local temple/religious festivals. A majority of these reasons relate to importance of social/kinship relations, as emphasized in Breza and Kaur (2025), but it is worth noting a large fraction of absenteeism attributable to low labor demand. This matches the rate of all-day firm closures recorded in the data - 10% of firms were closed for the entire day. 40% of closures were because the firm did not have work, followed by 36% because of local temple festivals or the firm owner's personal commitments, and 10% because of power cuts.

of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 129 minutes on average at baseline – treatment reduces dispersion by 22 minutes, a 17% reduction in dispersion, or improvement in coordination relative to control firms (column (1)). A more conservative measure that excludes extreme outliers - the 10-90 gap in arrival times – also reduces by 9.5 minutes (p-value $<$ 0.10) in Treatment firms, an 11% decrease relative to the control mean of 84 minutes. Finally, I look at treatment effects on deviation from the owner's preferred start time for the 90th percentile worker. Note that given the average size of firms (6 tailors), and assuming 71% attendance, the 90th percentile worker is equivalent to the last or second to last worker to arrive at the firm. This measure also tests whether these effects are driven by workers for whom the incentive was binding. The deviation of the 90th percentile worker reduces by 14 minutes (p-value $<$ 0.01) in treatment firms, relative to the control group mean of 47 minutes pre-incentives. These effects amount to a 30% reduction in dispersion, or improvement in coordination as a result of treatment.

5.2 Worker Output, Output per Hour, and Earnings

Next, I examine the effect of incentives on worker output, output per hour and earnings. Worker output is measured as the total quantity of pieces stitched by the worker in a week, across all tasks (and products). This measure does not necessarily correspond to firm output (total number of finished pieces), since a worker who stitched 100 collars and 100 sleeves would be considered to have worked on 200 pieces on that day. However, this measure aligns best with workers earnings, which are constructed as the product of the task specific piece rate and quantity produced on that task, summed over all tasks and products worked on in a week. Output and earnings are coded as 0 on days that a worker was absent, before aggregating to the weekly level. Since firms produce different types of products, I normalize weekly worker output to pre-incentive, product-level means and standard deviations to account for any level differences in task speed and production quantity across product types. I scale worker output by weekly hours worked to measure worker productivity among workers present at least 1 day of the week. For these outcomes, I pool "Late" and "Very Late" workers.

Worker Output Results for weekly worker output are reported in columns (1)-(2) of Table 4. The treatment effect on weekly worker output is 0.26 standard deviations and statistically significant at the 5% level. This amounts to a 24% increase in weekly output relative to the control group mean (column (1) of Table 4). This increase is driven equally by "Early" and "Late" workers (column (2)). In levels, this amounts to 900 additional pieces per

week in treatment firms over an average of 3,700 pieces per week in control firms. Notably, unlike the results on timely arrival, increases in worker output are not only concentrated among "Late" workers.

Worker output per hour Increases in worker output are not driven by longer work hours alone. Columns (5)–(6) of Table 4 report effects on output per hour. Output per hour rises by 0.007 SD for the average treatment worker (column 5, significant at the 10% level), and the effect is similar for "Early" and "Late" workers (column 6). The estimate is small, so I interpret it as suggestive evidence of higher per-hour productivity.

Output per hour rises as much for "Late" workers as for "Early" workers, even though "Late" workers drive the increase in work hours Table 3. "Late" workers therefore make more productive use of their time at the firm. Consistent with this, "Late" workers in treatment firms are 4 percentage points (25%) less likely to perform non-production tasks on days they are present (column 1, panel B of Table A9), although the estimate is not statistically significant. This reduction does not reflect higher helper attendance: helper attendance does not change in treatment firms (Table A10). Owners report that tailors take on helper tasks when orders are urgent (50%), helpers are absent (33%), there is no other work (30%), or tailors are idle (14%); owners could select multiple reasons. At endline, treatment firm owners are also more likely than control owners to report less idle time among tailors and helpers (Figure A7). Together, these results suggest better time allocation among "Late" workers in treatment firms.

Worker Earnings Results for weekly worker earnings *excluding* incentive payments are reported in columns (3)-(4) of Table 4. I report total worker earnings over the week by summing the product of task-specific output and piece rates over the week, assuming that piece rates account for level differences across products. The treatment effect on worker earnings mirrors the results on worker output, albeit smaller in magnitude and not statistically significant. Weekly earnings increase by INR 310 (9% of the control mean) for the average worker. The effects are similar across "Early" and "Late" workers: there is no differential effect for "Late" workers, and the total effect for "Late" workers is also not statistically significant.[20] Despite not being statistically significant, a 9% increase in earnings is an economically meaningful magnitude. Figure 8 shows a clear rightward shift in the distribution

[20] *Including* the treatment incentive payments, all workers in treatment firms earn INR 817 per week (see Table A6), and this effect is not statistically different for "Late" workers. These results suggest that if workers were to, hypothetically, arrive on time *without* any monetary incentives, they could have earned on average INR 310 more per week than their status quo arrival times. Depending on whether workers internalize indirect effects of timely arrival along the production line, the implied willingness to pay ranges between INR 310 to INR 817 in weekly earnings.

of change in earnings for workers in treatment firms, and this is not driven by outliers.

5.3 Evidence for coordination gains along the production line

The results on worker output, output per hour and earnings are experienced by all workers in the firm, despite the labor supply response driven by "Late" workers only, indicating the presence of positive spillovers within the firm as a result of treatment. Do these spillovers reflect gains along the production line? To assess this further, I examine whether spillover effects on "Early" workers are concentrated among Bottleneck firms – firms where "Late" workers performed upstream tasks at baseline. If the increase in worker output reflects gains from coordination along the production line, spillovers to workers for whom the incentives were not binding – "Early" workers – should appear only in firms where late arrivals actually held up production. Table 5 replicates results on worker output from Table 4 separately by bottleneck status. Panel A reports the treatment effect for the average worker, panel B reports heterogeneity by "Early" and "Late" workers, and column (1) replicates the spillover result for worker output from section 5.2. Columns (2) and (3) report results for for 52 Bottleneck firms and 40 Non-Bottleneck firms. The overall effect on weekly output, reported in column (2) of panel A, is larger in Bottleneck firms – 0.36 standard deviations (32% of the control mean, significant at the 1% level) in Bottleneck firms, compared to 0.12 standard deviations (12%, not statistically significant) in Non-Bottleneck firms (column (3) of panel A). However, I fail to reject the difference between these two samples. More importantly, which workers gain from the incentives differs across the two samples. Column (2) of panel B reflects the same pattern as column (2) of Table 4 – all workers, *including* "Early" workers gain from the treatment. "Early" workers experience a 0.39 SD increase in weekly output (equivalent to 1,000 pieces and significant at the 5% level), and the effect for "Late" workers is similar in magnitude (0.34 SD) and not statistically distinguishable from the effect for "Early" workers. By contrast in Non-Bottleneck firms (column (3), Panel B), the effect is concentrated entirely among "Late" workers, who experience an additional 0.40 SD increase in weekly output, $\approx$ 1,400 additional pieces (although the total effect is 0.34 SD, with a p-value of 0.11). Weekly worker earnings and output per hour (columns (2) and (3) in Table A7 and Table A8 respectively) exhibit the same pattern – all workers in Bottleneck firms experience an increase in earnings, although the estimate is not precisely estimated.

These patterns replicate for the alternate definition of hold-ups within the firm. Columns (4) and (7) Table 5 report results for Complex Task and No Complex Task firms respectively. Similar to the result for Bottleneck firms, spillovers from "Late" to "Early" are only present in Complex Task firms, where the firm is not able to cover for late workers (who work in

Simple Tasks): "Early" workers in Complex Task firms produce 0.40 SD more in weekly output (significant at 5% level), with no differential effect for "Late" workers (column (4), Table 5B). In contracts, the effect in No Complex Task firms is driven by "Late" workers, who produce 0.19 SD more in weekly output (not significant), while there is no effect for "Early" workers. The same pattern holds among firms where Complex tasks are downstream (where lateness on Simple tasks occurs upstream), reinforcing the importance of the sequential nature of tasks in generating these spillovers.

Taken together, spillovers to workers for whom the incentives were not binding appear only in firms where an upstream worker was late at baseline. This is what a coordination mechanism predicts, and is difficult to reconcile with alternative explanations for the output gains, such as peer effects from arriving together, or morning hours simply being more productive – both would generate gains among "Early" workers in Bottleneck and Non-Bottleneck firms alike.

5.4 Total Firm Output

Finally, how do the changes in individual worker output add up to total firm output and revenues? I construct total firm output using daily production data collected from workers by aggregating total quantity produced in two ways: (1) across *all* tasks, and (2) for the last task in the production line. These measures correspond to total units stitched in the firm, and total units of the finished garment respectively. Firm revenues and total labor costs are computed as the product of piece rates earned by owners and workers, and the total quantity of the last task in the production line. Both measures aggregate output across all production workers (i.e., including production by tailors, firm owners and managers, helpers and temporary workers). Treatment effects are estimated using Equation 2 and reported in Panel A of Table 6.

Total firm output increases by 0.30 standard deviations (column (1)) – this is equivalent to 6,800 additional pieces stitched at the firm, a 29% increase relative to the control mean and is statistically significant at the 10% level. The last task measure, reported in column (2) increases by 0.21 standard deviations, equivalent to 700 additional pieces of finished output. This measure is noisier, perhaps because the last task is under-reported if orders are still under production. Panel A of Table A15 reports these estimates after dropping the last week of data collection, and the magnitudes across both measures are similar. Reassuringly, a third measure counting all but the last task on the production line (column (3) of Table A15A) produces similar estimates. Finally, consistent with results on worker output, all of the gains

in total firm output are larger in Bottleneck firms, as reported in Column (1) of Table 6B. Total weekly output increased by 0.46 SD (41% relative to the control mean) in Bottleneck firms, while the effect in Non-Bottleneck firms is 0.13 SD (15% relative to the control mean). However, I fail to reject the difference between the two samples.

An alternate measure of total firm output is the quantity of new orders received by firms, reported in column (3) of Table 6. This measure was collected directly by firm owners and serves both as an independent measure of firm output[21], and firms' ability to procure new orders. Treatment firms procure 0.27 standard deviations more per week in new orders, a 21% increase relative to the control mean ($\approx$ 880 pieces). This measure is noisily estimated, but the magnitudes match those from daily production data collected from workers. Unlike the results on total firm output, both Bottleneck and Non-Bottleneck firms receive a larger quantity of new orders. However, both estimates are noisy and I am unable to reject the difference between both samples, making these results suggestive.

Finally, I find that treatment firms are more likely to deliver orders on time relative to control firms. Figure 10 reports treatment effects on order completion delays, defined as an indicator of whether the difference between planned and actual days of delivery exceeds 1 day. Treatment firms complete orders with significantly lower delays – the probability of completing an order after the planned delivery date falls by 10 percentage points, a 40% reduction relative to the control mean. Better coordination in worker arrival times facilitates timely completion of orders. In a context where most firms rely on repeat relationships with clients, timely completion is crucial for sustaining these relationships.

5.5 Profitability of Incentives

Total weekly firm revenues (column (1) of Table 7) in treatment firms increased by INR 3,257, an 8% increase relative to the control mean (INR 39,000), and total labor costs, *excluding* the incentives increase by INR 1,640, a 9% increase relative to the control mean (INR 17,843). However, both effects are noisy, and not statistically significant at conventional levels. Finally, total labor costs including the incentives paid to workers (column (3)) increase by INR 4,239 (24% relative to control firms).

These results suggest that improving worker coordination increases firm output, increases both firm revenues and labor costs (although noisily estimated). However, increases in labor costs including incentives outweigh the gain in revenues, resulting in a reduction in profits by INR 981 (5% of the control mean, reported in column (6)). Even including incentives paid

[21] 98% new orders were delivered, so this indicates total pieces delivered by the firm as well.

to the marginal workers only (i.e., "Late" and "Very Late" workers) – while not a feasible contract – firms would just break even (column (7)). Thus, while the lack of coordination and lateness in worker arrival times results in large loss in production, it is even costlier for firms to incentivize timely arrival. However, these calculations don't rule out the existence of any contract that could incentivize timely arrival. The choice of 20% daily bonus in this paper was deliberate in maximizing the first stage response from workers.

5.6 Why were owners not incentivizing timely arrival before the experiment?

A natural question arises – if lateness is indeed harmful to firm productivity, why do owners allow lateness in the status quo? I provide some suggestive reasons based on survey responses and anecdotes from the field.

First, the existing equilibrium in which these firms operate makes it hard for a single firm to enforce common work start times without raising wages or incentivizing workers. Firms are located in dense clusters, where over 95% of firms pay workers on piece rate contracts, and 80% owners report being okay with workers arriving 30 or more minutes late. When asked about the consequences of enforcing common work start times without incentives, owners believe that workers may leave the firm/refuse to work at the firm (52%), or they may be less willing to work extra hours when there are urgent orders (38%), and that it may harm their relationship with workers (31%). These reasons point to a combination of hiring frictions, idiosyncratic nature of demand and reputation concerns for owners.

Second, owners may have incorrect beliefs about workers' response to such incentives, or incorrect beliefs about the profitability of such incentives. In endline surveys, treatment firm owners are more likely to agree that their firm's total production and profits increased during the incentive weeks. This suggests that the experiment itself may have corrected owners' beliefs about the viability of such incentives. 76% of treatment owners reported that they would be willing to continue the incentives if the research team continued to pay for them, indicating that most owners found the incentives were useful. However, only 26% reported that they would be willing to continue the incentives by paying themselves. On average, treatment owners say that they would be willing to contribute INR 30 per day out of INR 100 for such incentives, and 19 out of 40 owners reported 0, indicating a low willingness to pay.

Finally, it may in fact be too costly to incentivize workers in the long-run. While this paper demonstrates the possibility of gains in the short-run, it is not clear whether workers

may continue to arrive on time if incentives were to continue indefinitely. The long-run response of workers depends on several other factors. In particular, missing or thin markets for services such as childcare, transportation (public transit, school buses), or the quality of public infrastructure (municipal water supply, regularity of government ration shops) affect workers' time-use outside of work, and are a key determinant of long-run responses. Perhaps low reported willingness to pay of the owners reflects their beliefs about long-run responses from workers.

Another important long-run consideration is contracting frictions. 7 out of 40 treatment firm owners said that they would be willing to contribute the full amount of the incentive that I offered. However, in conversations with owners after the endline survey, many expressed concern about being able to implement such incentives themselves. Their reasons ranged from contracting frictions to concerns about fairness within the firm. For example, one owner said: "*I'm willing to offer up to INR. 200, but if workers get used to it, they may start considering it as part of their salary and still not come on time.*" Another said "*workers did not feel upset because the incentive came from a third party. If I implement the same system, workers would perceive it as partiality.*" The existence of contracting frictions in developing countries is well-known, and is key to understanding why owners were not implementing such incentives before the experiment.

6 Conclusion

This paper experimentally tests whether irregular labor supply – specifically, a lack of coordination in worker arrival times – affects firm productivity in the context of a garment manufacturing cluster in India. I document that workers exercise considerable flexibility in when they arrive at work – at baseline, the 90-10 gap in worker arrival times is over 2 hours. This stands in contrast to firm owners' preference to have workers arrive at a common time, given the production line based organization of firms.

A daily bonus equivalent to 20% of worker earnings successfully increases timely arrival in Treatment firms by 43%, resulting in a 17% decrease in dispersion in arrival times at the firm level. This is a meaningful effect given the small size of these firms – incentives increase the fraction of workers who arrive within the first 15 minutes of opening time from 60% (3 out of 5 workers) to 80% (4 out of 5 workers).

As a result of the incentives, weekly output per worker increases by 24% for the average worker. This partly reflects an increase in productivity (output per worker-hour), given the smaller increase in weekly work hours. More importantly, this increase is distributed equally

among "Early" and "Late" workers in treatment firms, indicating the presence of positive spillovers since the timely arrival response was driven entirely by "Late" workers. These spillovers are present in firms where lateness creates the largest hold-ups along the production line, thus reflecting gains from better coordination in production. In firms with little hold-ups along the production line, gains are realized through a reduction in order completion delays – an equally crucial margin that helps firm owners maintain important client relationships. These gains translate almost one-to-one to total firm output, which increases by 29%. Average weekly revenues increase by 8% in treatment firms, and accounting for labor costs without incentives yields 8% higher profits. These results suggest that lateness does negatively affect firm productivity.

However, the costs involved in coordinating worker arrival times are too costly, and not profitable for firm owners. Perhaps not surprisingly, only 26% owners said they would be willing to offer such incentives themselves, and the average stated willingness to pay for a similar daily bonus was INR 30 – less than a third of the incentives that I offered in the experiment, indicating low willingness to pay. In a 3-12 month follow-up survey, I find no take-up of any kind of financial incentives by Treatment firms, and little effort overall in enforcing timely arrival.

Survey responses and anecdotal evidence suggest several possible constraints explaining why willingness to pay is low, and why owners were not implementing such incentives in the status quo. These reasons range from incorrect information and beliefs to norms, missing markets and contracting frictions that are prevalent in many low-middle income contexts. The latter three may require larger scale policy efforts to shift owners' willingness to change the types of contracts and incentives offered to workers. My results demonstrate the importance of policies aimed at improving supply of and access to childcare, and improving the quality of public services such as transportation and municipal residential services (e.g. water supply) as complementary to industrial policies. This also motivates the need for future research to understand what drives the nature of employment contracts offered by firms in developing countries, and their effects on firm productivity.

Finally, it is also striking that incentives did not change worker attendance, and that take-up does not increase to 100% workers arriving on time conditional on attendance. This is despite the incentives being 20% of average daily earnings, indicating that some workers very highly value flexibility. In other survey work with workers across Tirupur district, I document that the provision of flexibility varies systematically with firm size – small and medium firms are significantly more likely than large firms to offer their workers flexibility in work schedules. Consequently, there is sorting of workers into small ($\leq$ 20 workers) versus

large (> 20 workers) firms along gender, marital status and having children – characteristics that are positively correlated with demand for flexibility. While workers who work in small firms enjoy flexible work schedules, they do so at the cost of variable pay and lack of access to formal government benefits, such as state provided health insurance, social security, etc. To the extent that demand for flexibility affects workers' labor supply choices, such as whether to be self-employed, work as casual day labor (Oh et al., 2024; Cefala et al., 2024), or work in small firms without formal benefits, this can have important implications for the rate of wage-employment in developing countries. More research is needed to understand the drivers of the demand for flexibility in developing countries, as well as the broader economic implications on the functioning of labor markets and firms in such settings.

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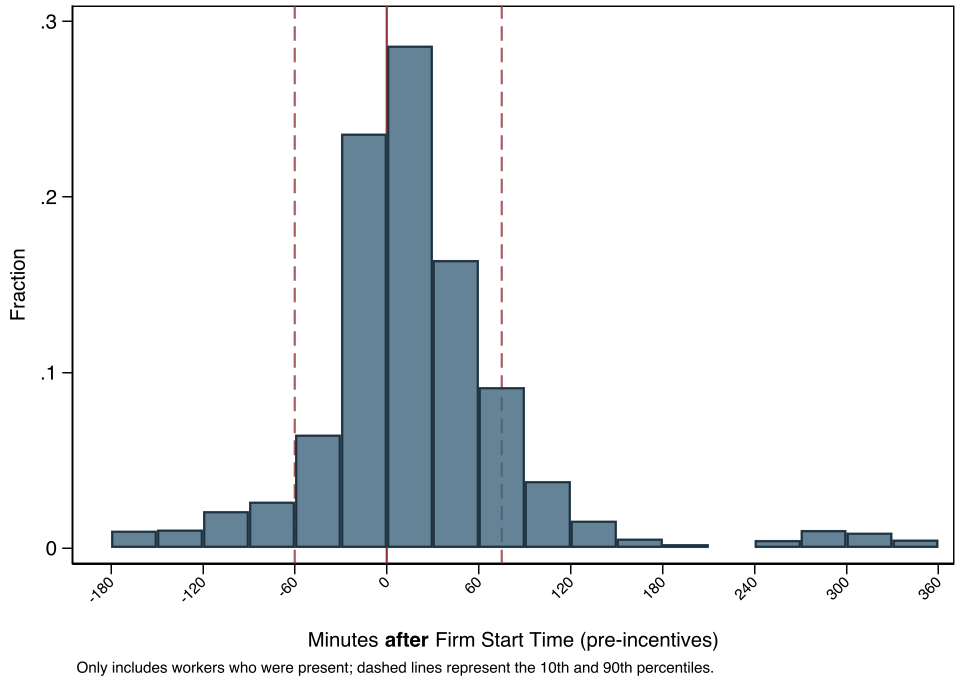
7 Figures

Figure 1: Stylized Production Line for T-Shirts



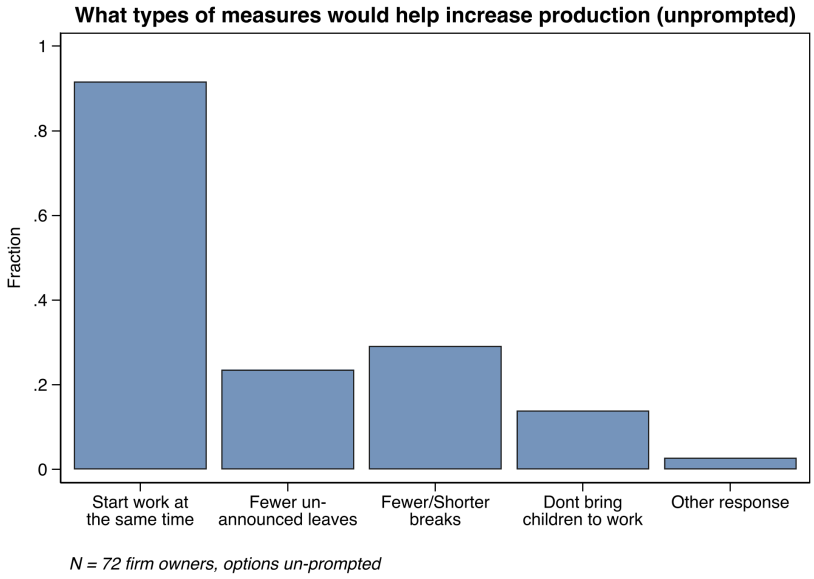
Notes: This figure presents a stylized example of a production line with 4 operations/tasks that make a T-Shirt. Firms receive cut panels of fabric which pass through a fixed sequence of stitching operations staffed by a distinct worker (tailor).

Figure 2: Dispersion in Worker Arrival Times, before Incentives

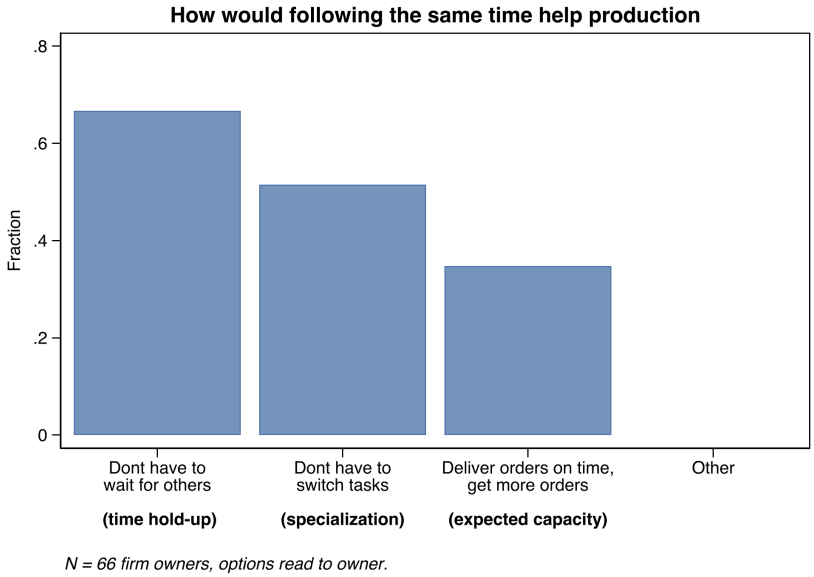


Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. N = 92 firms, ≈ 600 workers.

Figure 3: Importance of Common Arrival Times (Baseline)



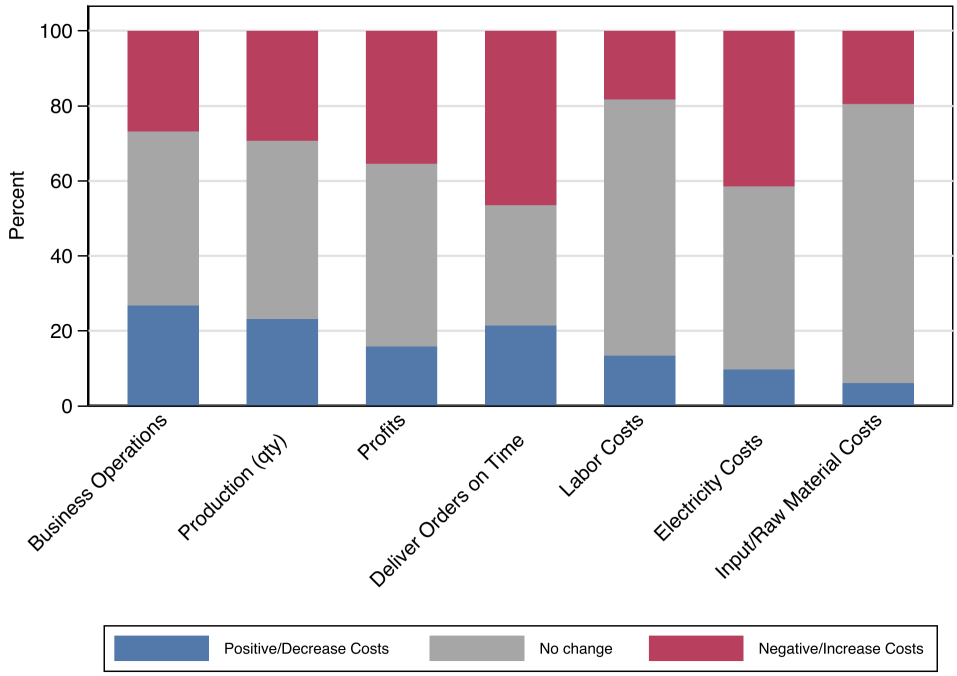
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "*What type of measures would help increase production?*", with options unprompted. N = 71. Panel B reports owner's responses from the question "*If everybody starts work at the same time, how will it improve production?*", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Helps me fulfill orders on time and get more orders, and 4 = Others.

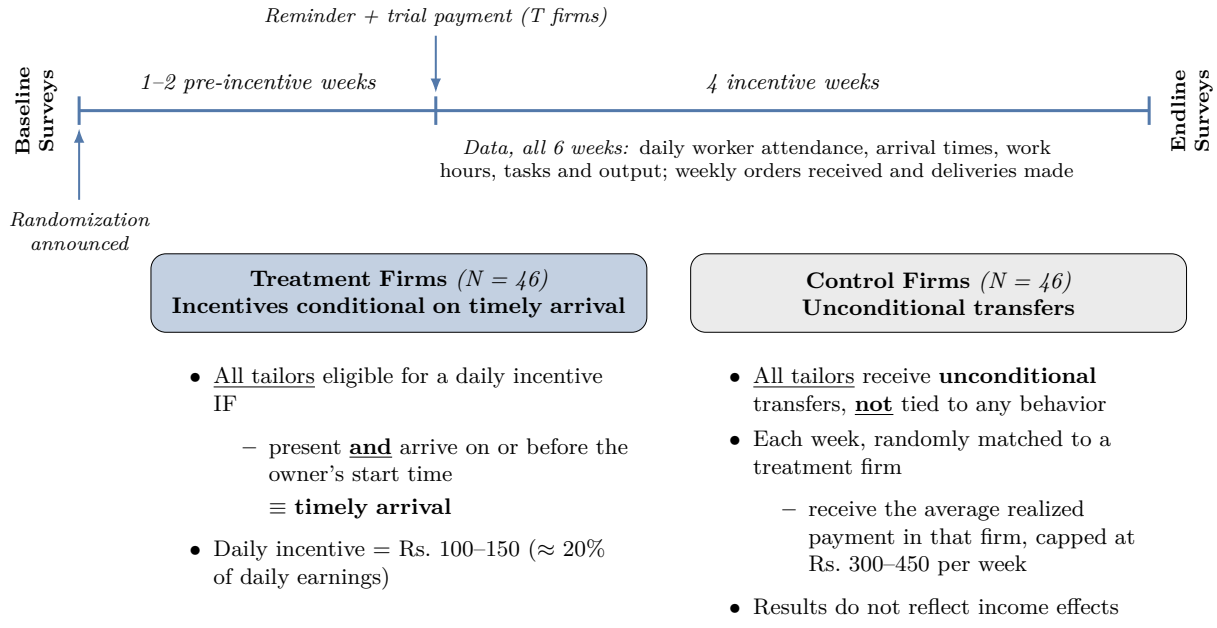
Figure 4: How Providing Flexibility Affects Firm Operations



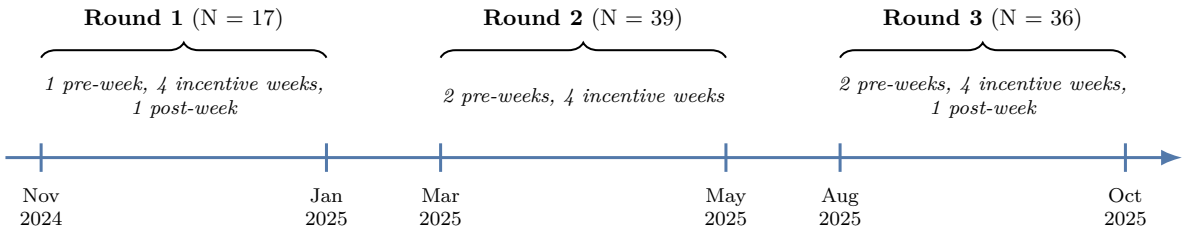
Notes: This figure reports owner's responses from the question: "*Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?*" This data is from the full sample of 92 firm owners.

Figure 5: Experimental Design and Implementation Timeline

Panel A: Experiment Design



Panel B: Implementation Timeline



Notes: Panel A shows the 6-week study window in each firm and the two treatment arms. Panel B shows the three implementation rounds; N is the number of firms in the analysis sample in each round.

Figure 6: Change in deviation from owner's preferred start time

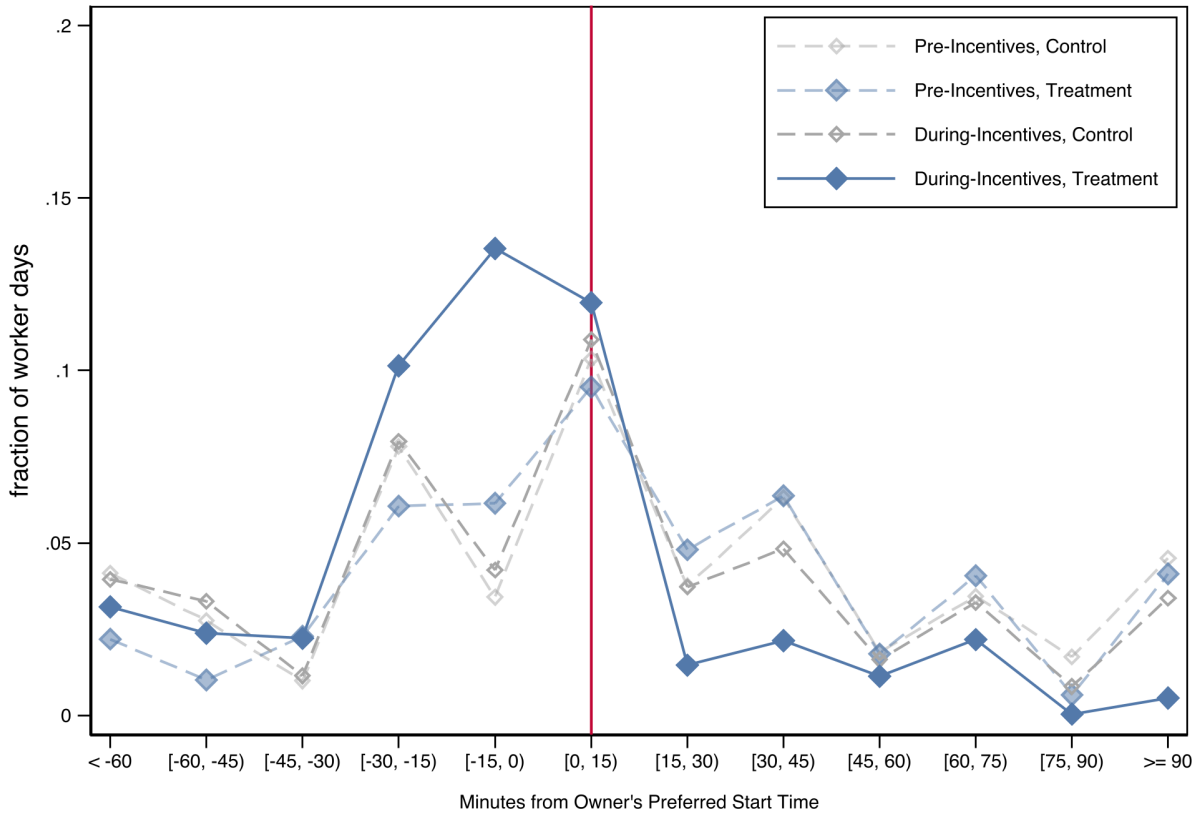


Figure 7: CDF of fraction of workers arriving in 15-minute intervals

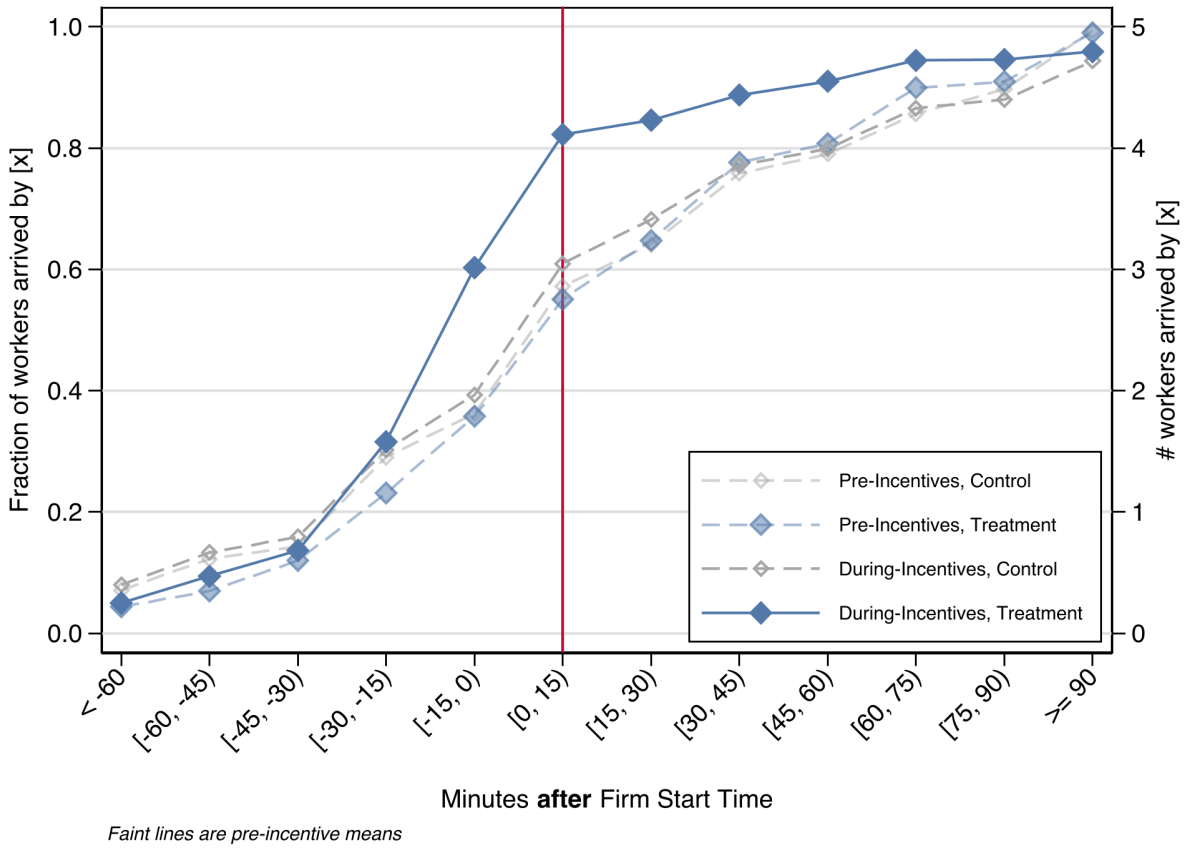
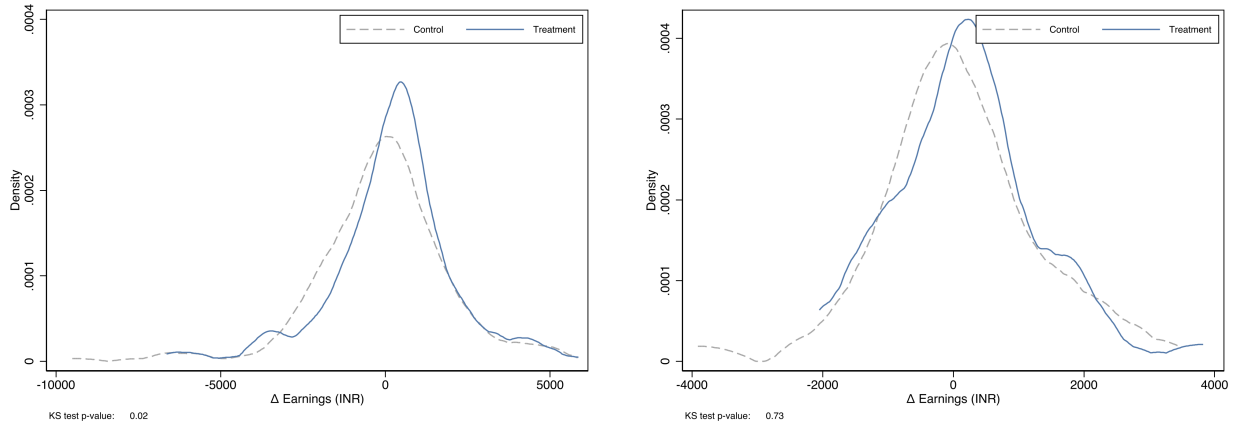


Figure 8: Distribution of the Change in Weekly Earnings

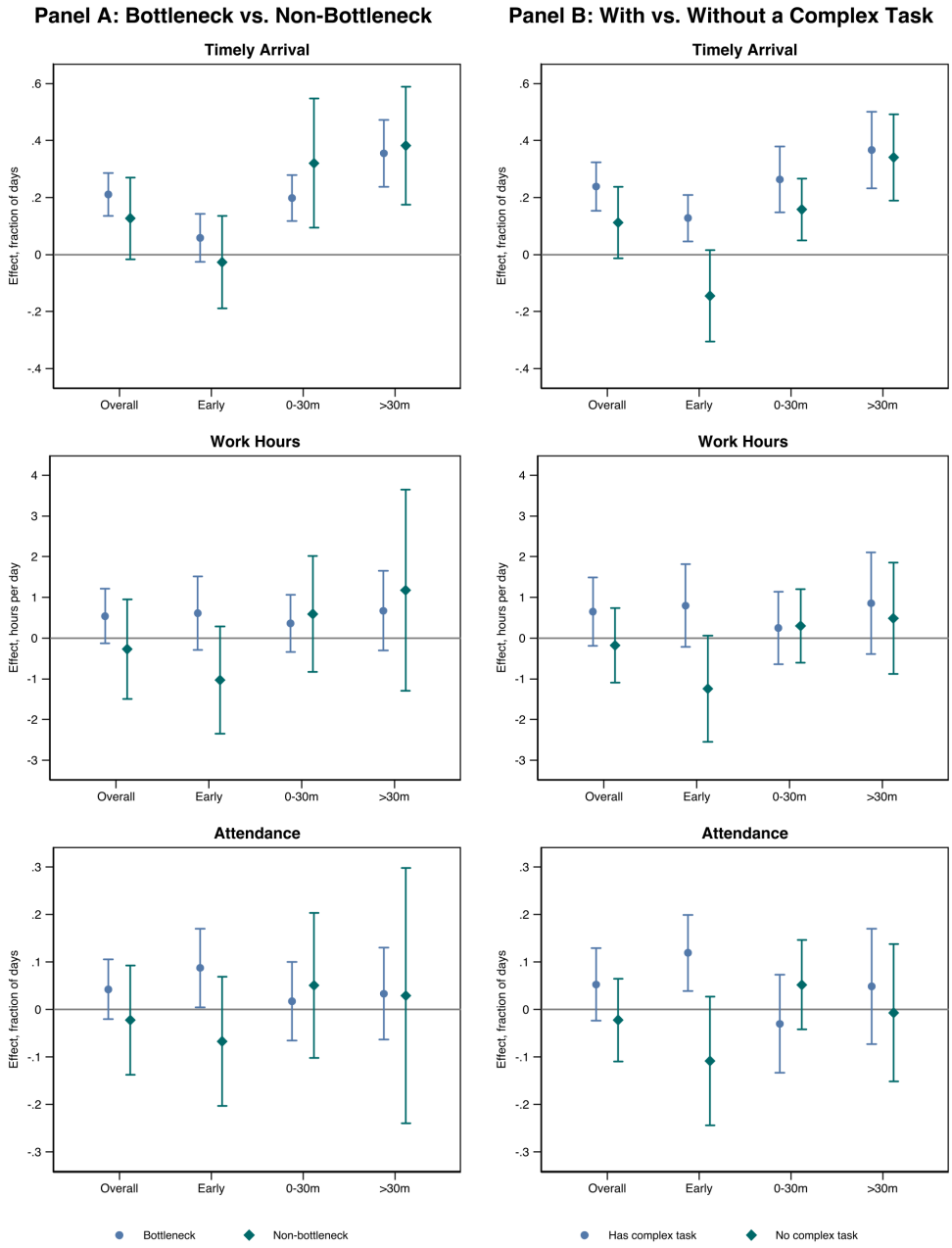


(a) Worker level

(b) Firm level

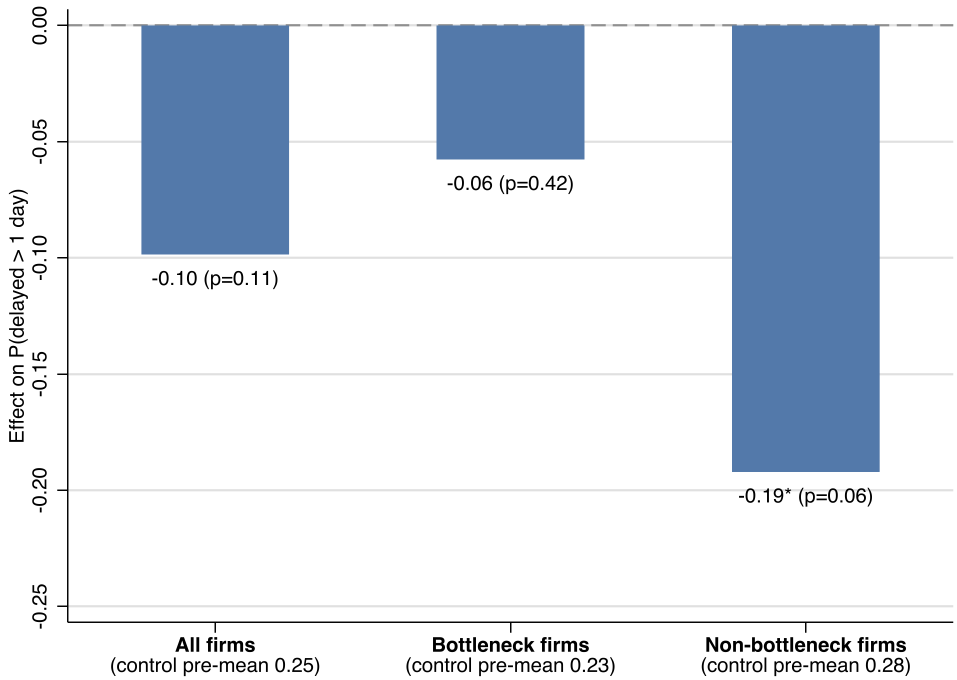
Notes: Kernel densities of the change in weekly earnings (INR, excluding incentives) between the pre-incentive and incentive weeks, for treatment (solid) and control (dashed) firms. Panel A: each worker's mean weekly earnings in the incentive weeks minus their mean in the pre-incentive weeks. Panel B: the firm average of the worker-level change. Earnings are coded as 0 for absent workers and winsorized at the 99th percentile. The KS test p-value is from a two-sample Kolmogorov–Smirnov test of equal treatment and control distributions.

Figure 9: Treatment Effects on Labor Supply by Bottleneck and Complex-Task Status



Notes: Each marker is the T × During coefficient from a separate regression on that group of firms, with a 95% confidence interval; standard errors are clustered at the firm level. Overall is the average effect; Early, 0–30m and >30m are the effects for workers who were early, 0–30 minutes late and more than 30 minutes late at baseline. Rows are outcomes. Panel A (left) splits firms by bottleneck status (at least one upstream worker late at baseline); Panel B (right) by whether the firm has any baseline complex (singer) task. Data at the worker-day level; all regressions include worker, day-of-week and round by week fixed effects.

Figure 10: Treatment Effects on Order Completion Delays



Notes: This figure plots treatment effects on the probability that an order is delivered more than 1 day after its planned delivery date, for all firms and separately for Bottleneck and Non-Bottleneck firms. Labels report point estimates and p-values; control means are pre-incentive. * $p < 0.10$.

8 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control	Treatment	T - C	*p-value*
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	6.47	6.22	-0.18	0.75
	(2.93)	(2.90)		
# Helpers	1.64	1.29	-0.37	0.12
	(1.15)	(1.15)		
Total # Machines	11.64	11.06	-0.44	0.59
	(4.86)	(3.63)		
Owner directly supervises workers	0.91	0.98	0.06	0.20
	(0.28)	(0.14)		
Firm Age (yrs)	6.68	6.76	0.05	0.97
	(5.29)	(6.18)		
Export orders	0.13	0.16	0.04	0.61
	(0.34)	(0.37)		
Only stitching section	0.57	0.63	0.05	0.59
	(0.50)	(0.49)		
Weekly Output	4075.61	4306.98	286.90	0.59
	(2320.43)	(2876.29)		
Revenues (last 30 days)	134444.19	126900.00	-8054.10	0.70
	(118877.68)	(101095.00)		
Profits (last 30 days)	33962.98	25916.67	-8173.26	0.36
	(58396.33)	(18038.77)		
Total Costs - wage bill (last 30 days)	32478.68	32978.37	842.78	0.83
	(20378.58)	(18684.70)		
Owner Demographics				
Age	39.21	37.80	-1.28	0.44
	(8.36)	(8.77)		
Male	0.85	0.82	-0.03	0.73
	(0.36)	(0.39)		
Completed Secondary School	0.57	0.55	-0.01	0.88
	(0.50)	(0.50)		
Commute Time (mins)	6.32	8.63	2.38	0.15
	(6.66)	(9.06)		
Experience in Industry (yrs)	19.15	18.35	-0.91	0.65
	(10.47)	(9.50)		
Owner does stiching work	0.87	0.88	0.00	1.00
	(0.34)	(0.33)		
Owner stitches as a regular tailor	0.43	0.43	-0.01	0.95
	(0.50)	(0.50)		
Observations	47	49		
Joint F-test p-value				0.475

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors. The joint F-test p-value is from a regression of treatment on all balance variables with strata fixed effects and robust SEs; missing values in weekly output, revenue, and profits are mean-imputed within strata for this test only.

Table 2: Balance on Baseline Worker Characteristics

	Control	Treatment	T - C	*p-value*
	Mean/SD	Mean/SD	Difference	
Age	37.22	36.49	-0.78	0.40
	(8.75)	(9.01)		
Male	0.45	0.43	-0.02	0.73
	(0.50)	(0.50)		
Married	0.87	0.84	-0.03	0.41
	(0.34)	(0.37)		
Has children $<$ 14 yrs	0.89	0.84	-0.05	0.72
	(1.26)	(1.16)		
Has elderly member in HH	0.36	0.40	0.04	0.41
	(0.48)	(0.49)		
Related to owner	0.21	0.21	0.00	0.99
	(0.41)	(0.41)		
Native to TN	0.92	0.95	0.03	0.24
	(0.27)	(0.21)		
Commute $<$ 15 mins	0.85	0.83	-0.02	0.48
	(0.36)	(0.38)		
Completed Secondary School	0.46	0.41	-0.05	0.26
	(0.50)	(0.49)		
Experience in Industry (yrs)	15.52	14.15	-1.43	0.14
	(9.25)	(9.53)		
Tenure at Firm (yrs)	2.57	2.38	-0.17	0.73
	(3.60)	(3.20)		
Observations	309	299		
Joint F-test p-value				0.340

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level. The joint F-test p-value is from a regression of treatment on all balance variables with strata fixed effects and SEs clustered at the firm level.

Table 3: Treatment Effects on Labor Supply: Overall and by Baseline Lateness

	Timely Arrival		**Work Hours**		**Attendance**	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.18***	0.01	0.28	-0.05	0.02	0.02
	[0.03]	[0.04]	[0.29]	[0.42]	[0.03]	[0.04]
Treatment × During × Late		0.21***		0.42		-0.01
		[0.05]		[0.47]		[0.05]
Treatment × During × Very Late		0.35***		0.78		0.00
		[0.06]		[0.51]		[0.05]
Control Mean (Pre-Incentives)	.41	.41	6.45	6.45	.71	.71
Omitted Group Mean (Pre-Incentives)		.69		7.42		.78
Total Effect (T × During + Inter.)		0.22, 0.36		0.37, 0.73		0.02, 0.03
p-val Total Effect = 0		0.00, 0.00		0.26, 0.12		0.65, 0.55
Treatment Effect %	43	54, 88	4	6, 11	3	2, 4
Worker, Week FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
N Worker-Days	17583	15852	16548	15272	17585	15853

Notes: All specifications include worker, day-of-week and round by week fixed effects. Standard errors are clustered at the firm level. Odd columns report the pooled treatment effect; even columns split it by the worker's average arrival delay in the pre-incentive weeks (Late = 0–30 minutes, Very Late = more than 30 minutes; the omitted group arrived on or before the owner's preferred start time). Attendance is defined as an indicator for whether a worker was present at the firm on that day. Timely arrival is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place. The lateness columns additionally drop workers with no measured baseline arrival time.
Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Weekly Output, Earnings and Output per Hour

	Weekly Output (SD)		**Weekly Earnings (INR)**		**Output per Hour (SD)**	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.26**	0.22	310.15	355.25	0.007*	0.007
	[0.11]	[0.14]	[271.34]	[370.18]	[0.004]	[0.004]
Treatment × During × Late		0.06		-72.57		-0.000
		[0.12]		[339.13]		[0.004]
Control Mean (Pre-Incentives)	3873.79	3873.79	3505.3	3505.3	92.1	92.1
Control SD (Pre-Incentives)	3582.97	3582.97	3073.26	3073.26	68.21	68.21
Omitted Group Mean (Pre-Incentives)		3847.45		4042.68		88.08
Total Effect (T × During + Inter.)		0.28		282.68		0.007
p-val Total Effect = 0		0.02		0.32		0.12
Treatment Effect %	24	26	9	8		
N Worker-Weeks	2663	2663	2663	2663	2398	2398

Notes: SEs clustered at firm level. All specifications include worker and round by week fixed effects. This regression excludes (a) owners/co-owners who also do stitching work at the firm, and (b) days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Earnings are coded as 0 for absent workers. Output and earnings are winsorized at the 99th percentile. Output per hour is normalized output divided by weekly work hours, on weeks the worker is present; its control mean and SD are in pieces per hour. Late pools all workers late at baseline. Total Effect reports the lincom of T × During plus the Late interaction.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Worker Output: Bottleneck and Complex-Task Splits

Weekly Worker Output (SD units)							
		Bottleneck		Complex Task			
	All Firms	**Bottleneck**	**Non-Bottleneck**	**Any Complex**	**Complex Downstream**	**Complex Upstream**	**No Complex**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Overall Treatment Effect							
Treatment × During	0.26**	0.36***	0.12	0.38**	0.48*	0.27	0.11
	[0.11]	[0.13]	[0.19]	[0.16]	[0.24]	[0.22]	[0.16]
p-val Bottleneck = Non-bottleneck		0.34					
p-val Any complex = No complex				0.26			
p-val Downstream = Upstream					0.48		
Panel B: Heterogeneity by Baseline Lateness							
Treatment × During	0.22	0.39**	-0.06	0.40**	0.51*	0.28	-0.04
	[0.14]	[0.19]	[0.19]	[0.19]	[0.28]	[0.26]	[0.19]
× Late	0.06	-0.05	0.40**	-0.04	-0.06	-0.01	0.23
	[0.12]	[0.17]	[0.17]	[0.18]	[0.27]	[0.24]	[0.16]
p-val Total Effect = 0	0.02	0.01	0.11	0.05	0.11	0.28	0.25
Control Mean (Pre-Incentives)	3873.79	3828.82	3933.42	3488.48	3441.48	3601.73	4320.85
Control SD (Pre-Incentives)	3582.97	3443.76	3768.8	3528.75	3636.15	3279.42	3616.53
N Worker-Weeks	2663	1728	935	1495	929	566	1153
N Firms	92	52	40	44	27	17	47

Notes: SEs clustered at firm level, in brackets. All specifications include worker and round by week fixed effects. The sample is workers (excluding owners/co-owners who also stitch) and excludes days of firm closure when no production took place. Output is normalized by product relative to pre-incentive week means and winsorized at the 99th percentile. Columns are estimation samples, each a separate regression. Bottleneck firms have at least one upstream worker late at baseline. Complex-task firms produce at least one baseline style with a complex (singer) task; in Complex Downstream firms the complex task follows a simple task in at least one baseline style, and Complex Upstream are the other complex-task firms. p-values test equal average effects from one regression with a separate T × During per group plus group × During. Panel B interacts T × During with *Late*, an indicator for workers late at baseline. Total Effect p-value tests T × During + Late interaction = 0.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Treatment Effects on Weekly Firm Output, New Orders and Order Delays

	Firm Output (SD) (worker-reported, daily)		**New Orders (SD)** (owner-reported, weekly)	**Delayed Order** (owner-reported, per order)
	All tasks	Last task		$\mathbb{1}\{\text{delay} > 1 \text{ day}\}$
	(1)	(2)	(3)	(4)
Panel A: All Firms				
Treatment × During	0.30*	0.21	0.27	-0.10
	[0.16]	[0.19]	[0.19]	[0.06]
Control Mean (Pre-Incentives)	23419.07	2785.89	4193.61	0.25
Control SD (Pre-Incentives)	22826.28	3381.37	3282.05	
p-val (T × D)	0.08	0.28	0.16	0.11
Treatment Effect %	29	25	21	-39
N Firm-Weeks	525	525	395	
N Orders				987
N Firms	92	92	87	88
Panel B: Bottleneck vs Non-Bottleneck Firms				
× Bottleneck	0.46**	0.29	0.29	-0.06
	[0.21]	[0.29]	[0.28]	[0.07]
× Non-bottleneck	0.13	0.11	0.23	-0.19*
	[0.27]	[0.21]	[0.22]	[0.10]
p-val Bottleneck = Non-bottleneck	0.35	0.62	0.88	0.28
N Firm-Weeks	525	525	391	
N Orders				981
N Firms	92	92	86	87

Notes: SEs in brackets, clustered at the firm level. Columns (1)–(3) are at the firm-week level and include firm and round by week FEs. Firm output counts pieces across all tasks on the production line (column 1) or at the last task, i.e. finished pieces (column 2). New orders are owner-reported material receipts, standardized to pre-incentive product-level means. Column (4) is at the order level: an order is delayed if delivered more than one day after its planned delivery date. Orders are indexed by planned delivery week, orders whose planned window straddles the start of the incentives are dropped, and the regression includes firm and round by planned-delivery-week FEs. Panel B interacts Treatment × During with an indicator for bottleneck firms (at least one upstream worker late at baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Treatment Effects on Weekly Firm Revenue, Labor Cost and Profit

		Labor Cost (INR)			**Profit (INR)**		
	Revenue (INR)		(+ incentives)	(+ incentives to late workers)		(− incentives)	(− incentives to late workers)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: All Firms							
Treatment × During	3258	1640	4239*	3105	1617	-981	153
	[4579]	[2301]	[2408]	[2349]	[2712]	[2687]	[2697]
Control Mean (Pre-Incentives)	39275.43	17843.57	17843.57	17843.57	21431.87	21431.87	21431.87
p-val (T × D)	0.48	0.48	0.08	0.19	0.55	0.72	0.95
Treatment Effect %	8	9	24	17	8	-5	1
N Firm-Weeks	525	525	525	525	525	525	525
N Firms	92	92	92	92	92	92	92
Panel B: Bottleneck vs Non-Bottleneck Firms							
× Bottleneck	4127	2360	5299	4188	1767	-1172	-61
	[6297]	[3404]	[3534]	[3449]	[3285]	[3251]	[3265]
× Non-bottleneck	1754	583	2540	1381	1172	-786	374
	[6553]	[2843]	[2982]	[2917]	[4510]	[4476]	[4499]
p-val Bottleneck = Non-bottleneck	0.79	0.69	0.55	0.53	0.92	0.95	0.94
N Firm-Weeks	525	525	525	525	525	525	525
N Firms	92	92	92	92	92	92	92

Notes: SEs in brackets, clustered at the firm level. Data at the firm-week level; all regressions include firm and round by week FEs. Revenue and labor cost are computed on finished pieces (the last task on the production line). Profit is revenue less labor cost. Columns (3) and (6) add the incentive payments made during the incentive period to labor cost and subtract them from profit; columns (4) and (7) count only the payments to workers late at baseline. Panel B interacts Treatment × During with an indicator for bottleneck firms (at least one upstream worker late at baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables and Figures

Table A1: Worker Labor Supply and Production Outcomes - Pre Incentives

		Avg. Dev. in Arrival Time before incentives					By Gender		
	All workers	≤ 0 (Early)	∈ (0,30] mins (Late)		> 30 mins (Very Late)		Male	Female	
	mean	mean	mean	p-value L-E	mean	p-value VL-E	mean	mean	p-value F-M
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Attendance, Arrival Times									
Present	0.73	0.78	0.83	0.64	0.75	0.02	0.72	0.74	0.46
Complied w preferred start time	0.41	0.70	0.44	0.00	0.08	0.00	0.54	0.31	0.00
Dev. in Arrival Time (mins)	7.90	-27.44	14.33	0.00	54.88	0.00	-13.51	24.95	0.00
Dev. in Exit Time (mins)	-37.08	-32.79	-36.34	0.82	-44.55	0.01	-31.70	-41.35	0.00
Daily Work Hours	6.64	7.55	7.56	0.03	6.12	0.00	6.93	6.41	0.00
Total # Breaks	1.85	2.10	2.05	0.30	1.75	0.00	1.89	1.81	0.03
Total Break Duration - mins	59.39	66.07	67.48	0.53	56.87	0.00	61.82	57.42	0.04
Production									
Avg Task Piece Rate (INR)	0.93	1.29	0.85	0.50	0.71	0.93	1.18	0.72	0.01
Output per hour	91.96	91.07	97.75	0.74	86.66	0.24	92.67	91.40	0.15
Earnings per hour (INR)	74.56	82.90	71.78	0.24	64.89	0.54	84.57	66.60	0.00
Daily Earnings (INR)	511.50	647.86	558.80	0.05	397.65	0.00	609.85	431.83	0.00
Upstream Task	0.24	0.28	0.26	0.89	0.21	0.07	0.22	0.25	0.94
Downstream Task	0.23	0.27	0.23	0.09	0.22	0.02	0.30	0.17	0.00
Complex Singer Task	0.08	0.17	0.05	0.07	0.01	0.00	0.16	0.02	0.00
Non-tailoring work	0.11	0.12	0.16	0.14	0.09	0.36	0.11	0.12	0.13
Demographics									
Female	0.55	0.32	0.60	0.00	0.84	0.00			
Married	0.87	0.82	0.88	0.05	0.95	0.00	0.81	0.93	0.00
Has children < 14 yrs	0.52	0.52	0.53	0.27	0.49	0.80	0.51	0.52	0.81
Commute < 15 mins	0.84	0.80	0.85	0.33	0.87	0.25	0.81	0.87	0.15
Tenure at Firm	2.85	2.74	3.32	0.72	2.79	0.55	3.12	2.64	0.16
Experience in Industry (yrs)	15.19	16.45	14.52	0.05	13.69	0.02	18.31	12.80	0.00
Owner Baseline Ranking									
Punctual	0.18	0.27	0.17	0.01	0.08	0.00	0.22	0.15	0.01
Reliable	0.19	0.21	0.21	0.47	0.15	0.04	0.25	0.15	0.00
Productive	0.18	0.24	0.16	0.02	0.13	0.00	0.21	0.16	0.07
Knowledgable about Prod.	0.18	0.23	0.16	0.02	0.14	0.13	0.25	0.13	0.00
benefit most from intervention	0.18	0.23	0.17	0.06	0.17	0.27	0.21	0.17	0.22
Observations	5084	1980	1399		1349		2277	2807	
Fraction of Workers		0.39	0.30		0.29		0.45	0.55	

Notes: Data at the worker-day level, includes 1-2 weeks of pre-incentives data, excludes days firm was closed. All outcomes coded as 0 if worker was absent. Piece rate and earnings are in INR.

Table A2: Treatment Effects on Deviation from Start Time

	Deviation from Start Time (mins)	
	(1)	(2)
Treatment × During	-11.18***	0.51
	[2.74]	[2.60]
Treatment × During × Late		-12.76***
		[2.74]
Treatment × During × Very Late		-27.59***
		[4.49]
Control Mean (Pre-Incentives)	5.59	5.59
Omitted Group Mean (Pre-Incentives)		-30.73
Total Effect (T × During + Inter.)		-12.25, -27.08
p-val Total Effect = 0		0.00, 0.00
Treatment Effect %	-200	-219, -484
Worker, Week FE	X	X
Day-of-week FE	X	X
N Worker-Days	12437	12308

Notes: All specifications include worker, day-of-week and round by week fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. The sample for these regressions excludes days on which a firm was shut, and no production took place. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Treatment Effects on Worker Exit Times

	$\mathbb{I}\{$**Left Before**$\}$				**Minutes after**	
	6.30 pm		**Closing Time**		**Regular Closing Time**	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.00	0.04	-0.01	-0.02	-7.30	-14.58
	[0.02]	[0.04]	[0.03]	[0.04]	[7.40]	[9.12]
Treatment × During × Late		-0.02		0.00		7.33
		[0.04]		[0.05]		[7.91]
Treatment × During × Very Late		-0.07**		0.03		17.60*
		[0.04]		[0.06]		[9.51]
Control Mean (Pre-Incentives)	.13	.13	.57	.57	-38.56	-38.56
Omitted Group Mean (Pre-Incentives)		.16		.63		-38.38
Total Effect (T × During + Inter.)		0.01, -0.04		-0.01, 0.01		-7.24, 3.02
p-val Total Effect = 0		0.66, 0.22		0.75, 0.78		0.32, 0.75
Treatment Effect %	2	11, -30	-2	-2, 3	19	19, -8
Worker, Week FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
N Worker-Days	17585	15853	17585	15853	12240	12112

Notes: All specifications include worker, day-of-week and round by week fixed effects. Standard errors are clustered at the firm level. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

Table A4: Treatment Effects on # and Duration of Breaks

	# Breaks		Total Break Duration (mins)	
Treatment × During	0.16	0.06	3.63	1.62
	[0.10]	[0.14]	[3.16]	[4.32]
Treatment × During × Late		0.07		1.16
		[0.17]		[5.29]
Treatment × During × Very Late		0.30*		6.93
		[0.15]		[5.32]
Control Mean (Pre-Incentives)	1.85	1.85	59.26	59.26
Omitted Group Mean (Pre-Incentives)		2.19		67.95
Total Effect (T × During + Inter.)		0.14, 0.36		2.78, 8.55
p-val Total Effect = 0		0.30, 0.02		0.49, 0.08
Treatment Effect %	9	7, 19	6	5, 14
Worker, Week FE	X	X	X	X
Day-of-week FE	X	X	X	X
N Worker-Days	17585	15853	17585	15853

Notes: Data at worker-day level. Outcomes coded 0 if worker was absent. Standard errors at clustered at the firm level. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker	**Mins between the 10th and 90th pctile worker**	**Deviation from start-time: 90th pctile worker**
	(1)	(2)	(3)
T × During	-22.07**	-9.59*	-13.96***
	[10.79]	[5.27]	[4.93]
Control Mean (Pre-Incentives)	128.96	83.93	47.11
Control Mean (Incentive Weeks)	115.32	70.44	31.25
Treatment Effect %	-17	-11	-30
Firm, Week FE	X	X	X
Day-of-week FE	X	X	X
N Firm-Days	2208	2949	2949

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include firm, day-of-week and round by week fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Weekly Worker Earnings

	Weekly Earnings (INR)			
	excl. Incentives		**incl. Incentives**	
	(1)	(2)	(3)	(4)
Treatment × During	310.15	355.25	817.37***	948.28**
	[271.34]	[370.18]	[276.26]	[376.17]
Treatment × During × Late		-72.57		-210.63
		[339.13]		[350.92]
Control Mean (Pre-Incentives)	3505.3	3505.3	3505.3	3505.3
Omitted Group Mean (Pre-Incentives)		4042.68		4042.68
Total Effect (T × During + Inter.)		282.68		737.64
p-val Total Effect = 0		0.32		0.01
Treatment Effect %	9	8	23	21
Worker, Week FE	X	X	X	X
N Worker-Weeks	2663	2663	2663	2663
N Firms	92	92	92	92

Notes: SEs clustered at firm level. All specifications include worker and round by week fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Earnings are coded as 0 for workers who are absent on a given day, and are winsorized at the 99th percentile. Late pools all workers late at baseline. Total Effect reports the lincom of T × During plus the listed interaction.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Worker Earnings: Bottleneck and Complex-Task Splits

Weekly Worker Earnings (INR)							
		Bottleneck		Complex Task			
	All Firms	**Bottleneck**	**Non-Bottleneck**	**Any Complex**	**Complex Downstream**	**Complex Upstream**	**No Complex**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Overall Treatment Effect							
Treatment × During	310.15	525.22	112.03	584.51	872.64	-15.60	-3.79
	[271.34]	[350.27]	[410.20]	[443.42]	[670.34]	[505.68]	[286.73]
p-val Bottleneck = Non-bottleneck		0.45					
p-val Any complex = No complex				0.28			
p-val Downstream = Upstream					0.26		
Panel B: Heterogeneity by Baseline Lateness							
Treatment × During	355.25	743.00	-240.79	785.33	1191.69	17.62	-239.51
	[370.18]	[514.17]	[493.38]	[575.52]	[818.22]	[738.23]	[339.99]
× Late	-72.57	-312.95	786.72*	-351.06	-565.05	-57.29	349.37
	[339.13]	[456.64]	[464.62]	[533.63]	[754.32]	[636.60]	[278.55]
p-val Total Effect = 0	0.32	0.22	0.18	0.35	0.38	0.93	0.71
Control Mean (Pre-Incentives)	3505.3	3946.21	2920.79	4206.35	4296.99	3988	2609.05
Control SD (Pre-Incentives)	3073.26	3515.53	2242.25	3641.05	3954.74	2756.95	1846.17
N Worker-Weeks	2663	1728	935	1495	929	566	1153
N Firms	92	52	40	44	27	17	47

Notes: SEs clustered at firm level, in brackets. All specifications include worker and round by week fixed effects. The sample is workers (excluding owners/co-owners who also stitch) and excludes days of firm closure when no production took place. Earnings exclude incentive payments, are coded as 0 for absent workers, and are winsorized at the 99th percentile. Columns are estimation samples, each a separate regression. Bottleneck firms have at least one upstream worker late at baseline. Complex-task firms produce at least one baseline style with a complex (singer) task; in Complex Downstream firms the complex task follows a simple task in at least one baseline style, and Complex Upstream are the other complex-task firms. p-values test equal average effects from one regression with a separate T × During per group plus group × During. Panel B interacts T × During with *Late*, an indicator for workers late at baseline. Total Effect p-value tests T × During + Late interaction = 0.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Worker Output per Hour: Bottleneck and Complex-Task Splits

Weekly Output per Hour (SD units)							
		Bottleneck		Complex Task			
	All Firms	Bottleneck	Non-Bottleneck	Any Complex	Complex Downstream	Complex Upstream	No Complex
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Overall Treatment Effect							
Treatment × During	0.007*	0.010**	0.005	0.011**	0.012	0.009*	0.000
	[0.004]	[0.004]	[0.007]	[0.005]	[0.008]	[0.005]	[0.006]
p-val Bottleneck = Non-bottleneck		0.64					
p-val Any complex = No complex				0.20			
p-val Downstream = Upstream					0.65		
Panel B: Heterogeneity by Baseline Lateness							
Treatment × During	0.007	0.011**	0.000	0.011*	0.015	0.006	-0.000
	[0.004]	[0.005]	[0.007]	[0.007]	[0.009]	[0.007]	[0.005]
× Late	-0.000	-0.002	0.010	-0.001	-0.006	0.004	0.001
	[0.004]	[0.004]	[0.010]	[0.006]	[0.009]	[0.010]	[0.005]
p-val Total Effect = 0	0.12	0.05	0.23	0.08	0.30	0.09	0.94
Control Mean (Pre-Incentives)	92.1	88.49	96.91	79.81	78.41	83.07	107.01
Control SD (Pre-Incentives)	68.21	61.82	75.82	65.23	68.48	57.28	69.15
N Worker-Weeks	2398	1570	828	1352	842	510	1031
N Firms	92	52	40	44	27	17	47

Notes: SEs clustered at firm level, in brackets. All specifications include worker and round by week fixed effects. The sample is workers (excluding owners/co-owners who also stitch) and excludes days of firm closure when no production took place. Output per hour is normalized output divided by weekly work hours, on weeks the worker is present; control mean and SD are in pieces per hour. Columns are estimation samples, each a separate regression. Bottleneck firms have at least one upstream worker late at baseline. Complex-task firms produce at least one baseline style with a complex (singer) task; in Complex Downstream firms the complex task follows a simple task in at least one baseline style, and Complex Upstream are the other complex-task firms. p-values test equal average effects from one regression with a separate T × During per group plus group × During. Panel B interacts T × During with *Late*, an indicator for workers late at baseline. Total Effect p-value tests T × During + Late interaction = 0.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Non-Production Tasks: Bottleneck and Complex-Task Splits

$\mathbb{1}${Did any non-production tasks at the firm}							
		Bottleneck		Complex Task			
	All Firms	Bottleneck	Non-Bottleneck	Any Complex	Complex Downstream	Complex Upstream	No Complex
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Overall Treatment Effect							
Treatment × During	-0.03	-0.04*	0.01	-0.01	0.00	-0.01	-0.04
	[0.03]	[0.03]	[0.06]	[0.04]	[0.04]	[0.08]	[0.05]
p-val Bottleneck = Non-bottleneck		0.39					
p-val Any complex = No complex				0.68			
p-val Downstream = Upstream					0.74		
Panel B: Heterogeneity by Baseline Lateness							
Treatment × During	-0.01	-0.04	0.01	0.01	-0.00	0.06	-0.04
	[0.04]	[0.03]	[0.07]	[0.05]	[0.04]	[0.10]	[0.06]
× Late	-0.03	-0.01	-0.01	-0.05	0.01	-0.13	0.00
	[0.03]	[0.03]	[0.09]	[0.04]	[0.04]	[0.08]	[0.04]
p-val Total Effect = 0	0.28	0.07	0.92	0.45	0.90	0.40	0.46
Control Mean (Pre-Incentives)	.16	.15	.17	.16	.16	.17	.15
Control SD (Pre-Incentives)	.36	.35	.38	.37	.36	.37	.36
N Worker-Days	12425	8145	4280	6956	4368	2588	5386
N Firms	92	52	40	44	27	17	47

Notes: SEs clustered at firm level, in brackets. All specifications include worker, day-of-week and round by week fixed effects. Data at the worker-day level, on days the worker is present; owners/co-owners who also stitch and days of firm closure are excluded. The outcome is an indicator for doing any non-production (helper) task or other work at the firm that day. Columns are estimation samples, each a separate regression. Bottleneck firms have at least one upstream worker late at baseline. Complex-task firms produce at least one baseline style with a complex (singer) task; in Complex Downstream firms the complex task follows a simple task in at least one baseline style, and Complex Upstream are the other complex-task firms. p-values test equal average effects from one regression with a separate T × During per group plus group × During. Panel B interacts T × During with *Late*, an indicator for workers late at baseline. Total Effect p-value tests T × During + Late interaction = 0.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Treatment Effects on Helper Attendance - Firm Level

	Fraction Helpers Present
	(1)
Treatment × During Incentives	0.03
	[0.04]
Control Mean (Pre-Incentives)	.83
Control Mean (Incentive Weeks)	.81
Treatment Effect %	4
Firm, Week FE	X
Day-of-week FE	X
N Firm-Days	3039

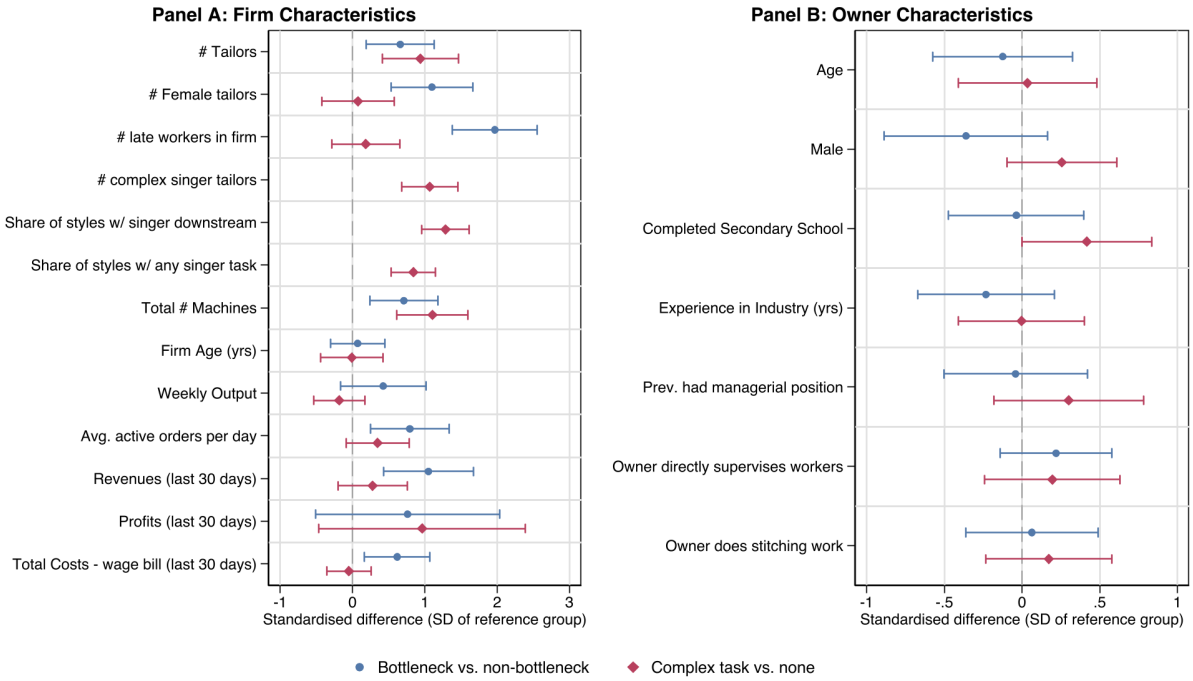
Notes: This table reports firm-day level outcomes on helper attendance. All specifications include firm, day-of-week and round by week fixed effects. Standard errors are clustered at the firm level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Worker Output and Earnings When an Upstream Worker Is Late

	All days			1 active order		
	Mid- and downstream	**Mid-stream**	**Down-stream**	**Mid- and downstream**	**Mid-stream**	**Down-stream**
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Total pieces stitched (qty.)						
≥1 upstream worker late	-35.87	-24.84	-51.21*	-121.48***	-105.33***	-157.19*
	[25.12]	[32.13]	[29.63]	[45.13]	[35.69]	[83.06]
Base mean (no upstream late)	981.72	946.63	1040.93	910.47	897.89	936.13
Worker FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
Firm-order FE	X	X	X	X	X	X
N	7460	4466	2970	2002	1198	806
Panel B: Total earnings (INR)						
	(1)	(2)	(3)	(4)	(5)	(6)
≥1 upstream worker late	-21.66	-2.11	-39.40	-76.11**	-86.72***	-44.45
	[20.45]	[25.96]	[34.99]	[33.11]	[28.11]	[55.34]
Base mean (no upstream late)	719.75	696.32	746.75	715.15	736	665.38
Worker FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
Firm-order FE	X	X	X	X	X	X
N	7460	4466	2970	2002	1198	806

Notes: Data at worker-order-day level, pre-incentive weeks + control firms. This table reports results from a regression of daily output and earnings for worker i, working on order o on day d on an indicator for whether another worker performing an upstream task on the same order arrived late that day. Columns 1 and 4 report results for both midstream and downstream workers, while remaining columns seperate the two samples. Cols. 1–3 use all such days; cols. 4–6 restrict to firm-days with one active order. Within each block, columns are downstream (non-early) tasks, mid tasks, and late tasks. All specs include worker, day-of-week, and firm-order fixed effects; SEs clustered at firm level, in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure A1: Baseline Firm and Owner Characteristics by Bottleneck and Complex-Task Status



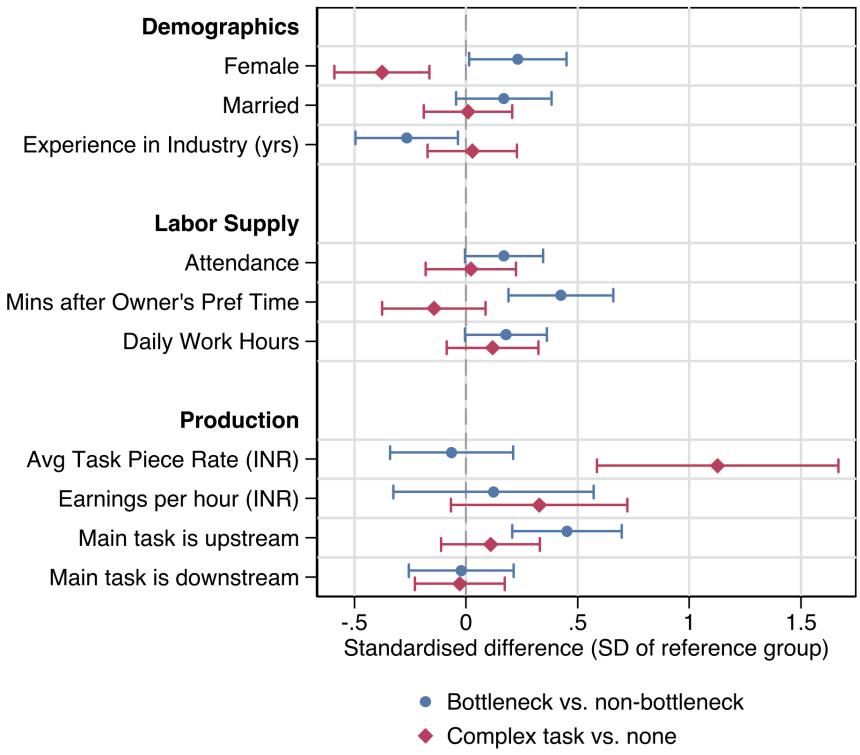
Notes: Each marker is the difference between the two groups of firms, from a regression of the characteristic on the group indicator with strata fixed effects and robust standard errors, divided by the SD of the reference group (non-bottleneck firms; firms without a complex task), with a 95% confidence interval. Bottleneck firms have at least one upstream worker late at baseline; complex-task firms produce at least one baseline style with a complex (singer) task.

Table A12: Baseline Firm and Owner Characteristics by Bottleneck and Complex-Task Status

	Bottleneck Status			Complex Task		
	Non-Bottleneck Mean/SD	Bottleneck Mean/SD	*p-value*	No Complex Mean/SD	Complex Mean/SD	*p-value*
Firm Characteristics						
# Tailors	5.70	6.88	0.01	5.26	7.64	0.00
	(2.71)	(3.05)		(2.30)	(3.09)	
# Helpers	1.15	1.73	0.00	1.53	1.41	0.51
	(1.10)	(1.17)		(1.25)	(1.11)	
Fraction female workers	0.53	0.64	0.16	0.67	0.50	0.02
	(0.27)	(0.24)		(0.26)	(0.23)	
# late workers in firm	1.57	4.17	0.00	2.81	3.30	0.45
	(1.43)	(2.12)		(2.11)	(2.42)	
Avg. active orders per day	1.94	2.24	0.01	2.02	2.22	0.12
	(0.61)	(0.88)		(0.71)	(0.86)	
Total # Machines	10.32	12.23	0.00	9.36	13.64	0.00
	(3.97)	(4.42)		(3.38)	(4.17)	
Owner directly supervises workers	0.90	0.98	0.24	0.94	0.95	0.38
	(0.30)	(0.14)		(0.25)	(0.21)	
Firm Age (yrs)	6.53	6.98	0.71	6.79	6.84	0.97
	(6.69)	(5.12)		(5.31)	(6.44)	
Export orders	0.10	0.19	0.23	0.17	0.14	0.39
	(0.30)	(0.40)		(0.38)	(0.35)	
Only stitching section	0.55	0.63	0.86	0.70	0.48	0.15
	(0.50)	(0.49)		(0.46)	(0.51)	
Weekly Output	3912.12	4545.83	0.16	4531.71	4000.00	0.31
	(2116.57)	(2886.94)		(3089.45)	(2013.11)	
Business Finances						
Revenues (last 30 days)	98783.78	156946.00	0.00	115471.11	150880.49	0.26
	(73445.42)	(126551.47)		(97378.91)	(123145.08)	
Profits (last 30 days)	26270.27	33548.16	0.24	23355.56	37888.00	0.19
	(16517.08)	(54894.03)		(13741.37)	(60321.80)	
Total Costs - wage bill (last 30 days)	28056.00	36690.35	0.01	32887.87	33602.45	0.75
	(20092.20)	(18849.50)		(24065.04)	(13775.41)	
Owner Demographics						
Age	39.80	38.21	0.59	38.55	39.57	0.88
	(8.87)	(8.16)		(7.94)	(8.93)	
Male	0.90	0.77	0.18	0.74	0.91	0.16
	(0.30)	(0.43)		(0.44)	(0.29)	
Completed Secondary School	0.60	0.58	0.87	0.45	0.73	0.05
	(0.50)	(0.50)		(0.50)	(0.45)	
Commute Time (mins)	7.70	7.25	0.77	6.85	8.14	0.69
	(7.80)	(8.46)		(8.43)	(7.95)	
Experience in Industry (yrs)	20.62	17.62	0.30	19.32	18.82	0.99
	(10.67)	(9.43)		(10.58)	(9.45)	
Owner does stitching work	0.85	0.88	0.77	0.87	0.86	0.41
	(0.36)	(0.32)		(0.34)	(0.35)	
Owner stitches as a regular tailor	0.42	0.44	0.59	0.55	0.32	0.12
	(0.50)	(0.50)		(0.50)	(0.47)	
Observations	40	52		47	44	

Notes: This table reports means and SDs of baseline firm and owner characteristics by firm group. Bottleneck firms have at least one upstream worker late at baseline; complex-task firms produce at least one baseline style with a complex (singer) task. p-values are from a regression of each variable on the group indicator with strata fixed effects. Robust standard errors.

Figure A2: Baseline Worker Characteristics by Bottleneck and Complex-Task Status



Notes: Each marker is the standardized difference between the two groups of firms (SD of the reference group), with a 95% confidence interval; standard errors clustered at the firm level. Demographics are from the worker baseline survey; labor supply and production are pre-incentive worker-day outcomes. Groups as in Figure A1.

Table A13: Baseline Worker Characteristics by Bottleneck and Complex-Task Status

	Bottleneck Status			**Complex Task**		
	Non-Bottleneck	Bottleneck	*p-value*	No Complex	Complex	*p-value*
	Mean/SD	Mean/SD		Mean/SD	Mean/SD	
Age	37.25	36.74	0.30	36.53	37.18	0.63
	(9.06)	(8.78)		(9.50)	(8.41)	
Male	0.50	0.40	0.11	0.34	0.52	0.00
	(0.50)	(0.49)		(0.48)	(0.50)	
Married	0.84	0.87	0.22	0.84	0.87	0.54
	(0.37)	(0.34)		(0.37)	(0.34)	
Has children < 14 yrs	0.84	0.90	0.74	0.89	0.87	0.78
	(0.96)	(1.35)		(1.50)	(0.95)	
Has elderly member in HH	0.39	0.38	0.71	0.40	0.37	0.58
	(0.49)	(0.49)		(0.49)	(0.48)	
Related to owner	0.16	0.24	0.33	0.26	0.17	0.17
	(0.37)	(0.43)		(0.44)	(0.38)	
Native to TN	0.92	0.95	0.31	0.98	0.91	0.00
	(0.27)	(0.22)		(0.15)	(0.29)	
Commute < 15 mins	0.84	0.84	0.79	0.86	0.82	0.30
	(0.37)	(0.37)		(0.35)	(0.38)	
Completed Secondary School	0.40	0.46	0.21	0.40	0.46	0.12
	(0.49)	(0.50)		(0.49)	(0.50)	
Experience in Industry (yrs)	15.51	14.44	0.03	14.46	15.13	0.73
	(9.80)	(9.24)		(10.08)	(8.99)	
Tenure at Firm (yrs)	2.39	2.53	0.90	3.09	1.99	0.01
	(3.78)	(3.21)		(4.02)	(2.83)	
Observations	223	368		257	330	

Notes: This table reports means and SDs of baseline worker characteristics by firm group. Bottleneck firms have at least one upstream worker late at baseline; complex-task firms produce at least one baseline style with a complex (singer) task. p-values are from a regression of each variable on the group indicator with strata fixed effects. Standard errors clustered at the firm level.

Table A14: Heterogeneity in Treatment Effects by Gender

	Labor Supply			Production		
	Timely Arrival	**Work Hours**	**Attendance**	**Weekly Output (SD)**	**Weekly Earnings (INR)**	**Output per Hour (SD)**
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.10**	0.07	0.01	0.25*	164.62	0.007
	[0.04]	[0.32]	[0.03]	[0.13]	[345.07]	[0.005]
Treatment × During × Female	0.13***	0.37	0.01	0.01	259.67	0.000
	[0.04]	[0.33]	[0.04]	[0.10]	[261.11]	[0.004]
Control Mean (Pre-Incentives)	.41	6.45	.71	3873.79	3505.3	92.1
Omitted Group Mean (Male, Pre-Incentives)	.55	6.76	.71	3818.15	4097.95	84.9
Total Effect (T × During + Female)	0.23	0.44	0.02	0.26	424.28	0.007
p-val Total Effect = 0	0.00	0.21	0.47	0.02	0.11	0.06
Treatment Effect %	57	7	3	24	12	
Unit of Observation	Worker-days	Worker-days	Worker-days	Worker-weeks	Worker-weeks	Worker-weeks
Observations	17583	16548	17585	2663	2663	2398

Notes: SEs clustered at firm level. Columns (1)–(3) are at the worker-day level and include worker, day-of-week and round by week fixed effects; columns (4)–(6) are at the worker-week level and include worker and round by week fixed effects, so observation counts are not comparable across the two sets. Timely arrival is an indicator for being present and arriving by the owner's preferred start time; attendance is an indicator for being present. Output is normalized by product relative to pre-incentive week means; earnings exclude incentive payments and are coded as 0 for absent workers; both are winsorized at the 99th percentile. Output per hour is normalized output divided by weekly work hours, on weeks the worker is present; its control and omitted-group means are in pieces per hour. Total Effect reports the lincom of T × During plus the Female interaction.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Treatment Effects on Weekly Firm Output and New Orders, Weeks 1–3

	Firm Output (SD) (worker-reported, daily)			**New Orders (SD)** (owner-reported, weekly)
	All tasks	Last task	All but last	
	(1)	(2)	(3)	(4)
Panel A: All Firms				
Treatment × During	0.33*	0.31	0.33*	0.26
	[0.17]	[0.20]	[0.18]	[0.19]
Control Mean (Pre-Incentives)	23419.07	2785.89	20633.18	4193.61
Control SD (Pre-Incentives)	22826.28	3381.37	20157.05	3282.05
p-val (T × D)	0.06	0.11	0.07	0.17
Treatment Effect %	32	38	32	20
N Firm-Weeks	434	434	434	334
N Firms	92	92	92	87
Panel B: Bottleneck vs Non-Bottleneck Firms				
× Bottleneck	0.46*	0.42	0.50**	0.30
	[0.24]	[0.30]	[0.24]	[0.27]
× Non-bottleneck	0.20	0.15	0.17	0.18
	[0.26]	[0.23]	[0.28]	[0.22]
p-val Bottleneck = Non-bottleneck	0.48	0.49	0.38	0.72
N Firm-Weeks	434	434	434	331
N Firms	92	92	92	86

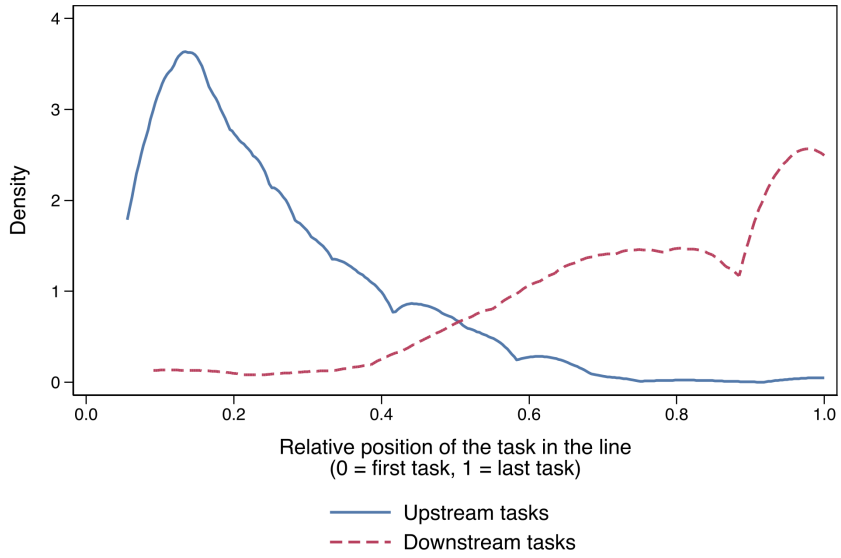
Notes: SEs in brackets, clustered at the firm level. Data at the firm-week level; all regressions include firm and round by week FEs. Firm output counts pieces across all tasks on the production line (column 1) or at the last task, i.e. finished pieces (column 2). Column (3) counts pieces at all tasks except the last. New orders are owner-reported material receipts, standardized to pre-incentive product-level means. Panel B interacts Treatment × During with an indicator for bottleneck firms (at least one upstream worker late at baseline). Restricted to weeks 1–3 of the incentive period. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A16: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Pay Structure				
Piece Rate	0.90	(0.30)	0.60	(0.49)
Shift Rate	0.06	(0.23)	0.40	(0.49)
Average Monthly Earnings (Rs.)	16087.43	(5899.93)	16713.75	(5323.63)
Work Timings				
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be <= 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

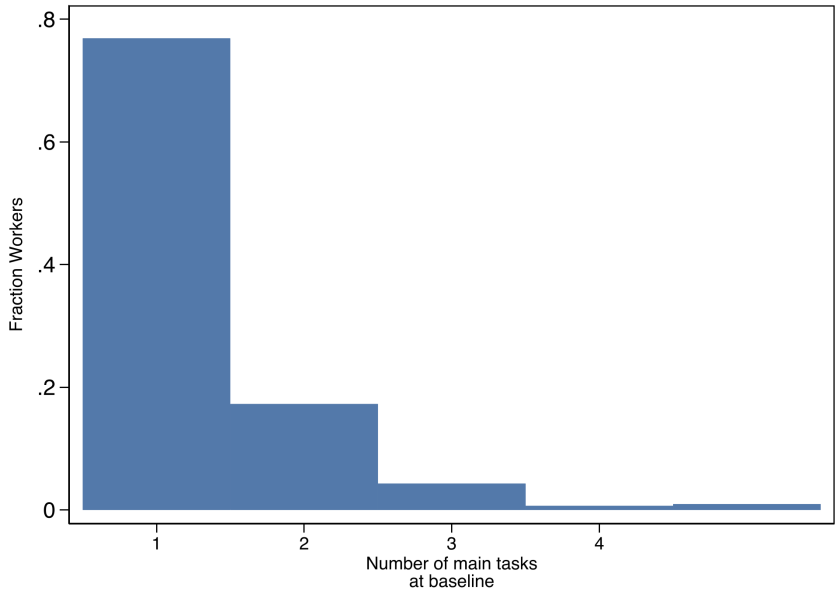
Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.

Figure A3: Position of Upstream and Downstream Cut-off Tasks in the Production Line



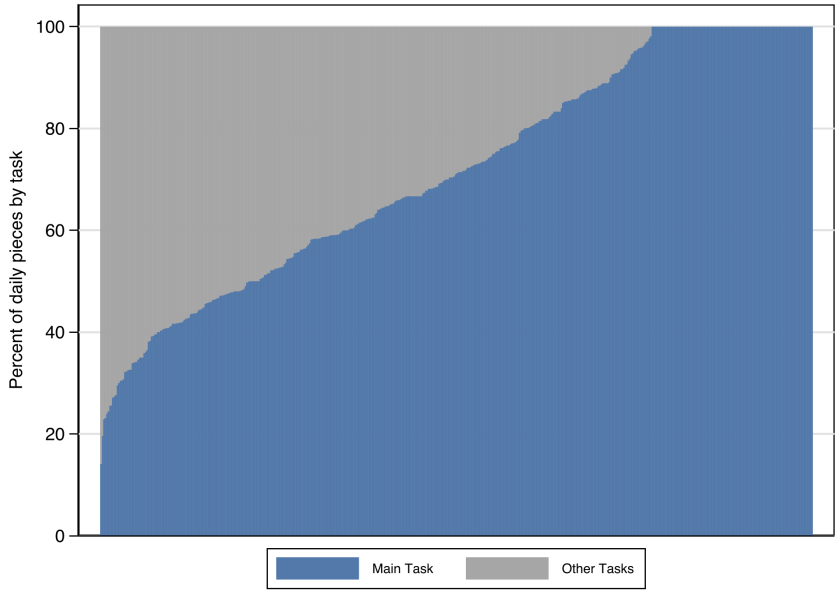
Notes: This figure plots the distribution of the relative position in the production line of tasks classified as upstream cut-off tasks (solid line) and downstream cut-off tasks (dashed line). Relative position is a task's order in the production line divided by the total number of tasks for that product (0 = first task, 1 = last task). Upstream cut-off tasks are tasks that must be completed before remaining tasks can proceed (e.g., attaching shoulders on a t-shirt); downstream cut-off tasks are tasks that must be completed for a garment to be finished (e.g., bottom hems on a t-shirt, or elastic bands on pants). Data from the baseline style rosters.

Figure A4: Number of Main Tasks per Worker



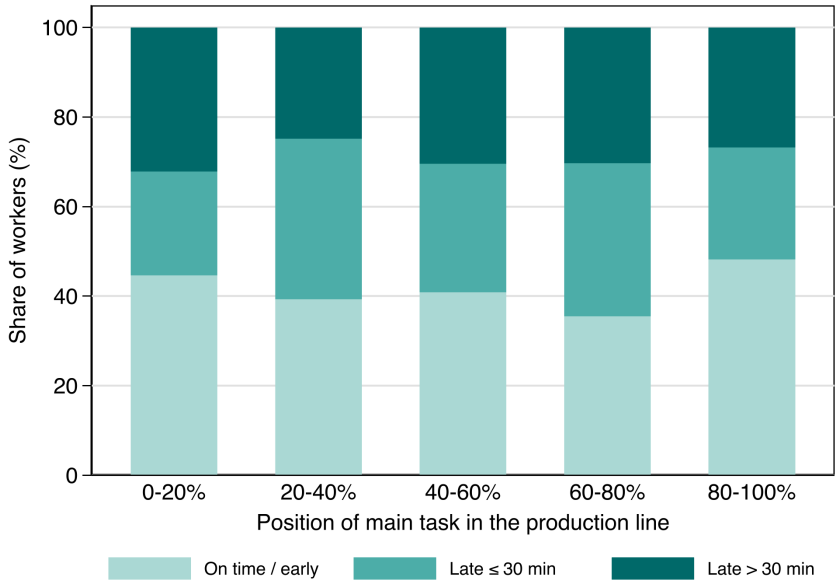
Notes: This figure shows the distribution of the number of main tasks per worker. For every worker, I define a "Main Task" as a task that I observe the worker performing every day she was present in a given pre-incentive week.

Figure A5: Task Specialization



Notes: Each bar is one worker. This figure shows the share of each worker's daily pieces from their main task and from all other tasks, averaged over pre-incentive days. A worker's main task is defined as the task that I observe that worker performing every day she was present in a given pre-incentive week. Workers with more than one main task have all main tasks counted together.

Figure A6: Baseline Lateness by Position of Main Task in the Production Line



Notes: This figure plots the share of "Early", "Late" and "Very Late" workers by the position of their main task in the production line. Task order is divided by 9, the median number of tasks per product, and capped at 1, so that relative task order denotes fifths of the production line.

Figure A7: Self-Reported Idle Time among Workers

